# FERTILIZATION

- PROCESS OF FUSION OF WELL DIFFERENTIATED HAPLOID GERM CELLS (OVUM AND SPERMATOZOON) RESULTING IN THE FORMATION OF MONONUCLEATED SINGLE ZYGOTE.
- OUT OF A FEW HUNDRED CAPACITATED SPERMS, ONLY ONE PIERCES THE ZONA PELLUCIDA AND ENTERS THE OVUM.
- THE NUCLEUS OF THE OVUM BECOMES FEMALE PRONUCLEUS. THE HEAD OF THE SPERMATOZOON (FORMED FROM THE NUCLEUS) SEPARATES FROM THE MIDDLE PIECE AND TAIL TRANSFORMING ITSELF INTO THE MALE PRONUCLEUS.
- THE 23 CHROMOSOMES OF THE FEMALE PRONUCLEUS AND 23 OF THE MALE PRONUCLEUS GET MIXED UP AND FORM 23 PAIRS (46 CHROMOSOMES). HENCE, FERTILIZATION IS THE FUSION OF MALE AND FEMALE PRONUCLEI/CHROMOSOMES.
- FERTILIZATION OCCURS IN THE AMPULLA OR LATERAL ONE-THIRD OF UTERINE TUBE

THREE STAGES CAN BE DESCRIBED:

1. APPROXIMATION OF GAMETES
2. CONTACT AND FUSION OF GAMETES
3. EFFECTS/RESULTS OF FERTILIZATION

APPROXIMATION OF GAMETES: BY TRANSPORT OF MALE AND FEMALE GAMETES IN THE FEMALE GENITAL TRACT

- SPERMATOZOON TRANSPORT
  - THE PROSTAGLANDINS PRESENT IN SEMEN STIMULATE PERISTALTIC CONTRACTIONS OF FEMALE GENITAL TRACT AT THE TIME OF SEXUAL INTERCOURSE.
  - THE SPERMATOZOA ARE ATTRACTED TO THE OVUM BY CHEMOTAXIS
  - CAPACITATION- FINAL STEP IN MATURATION OF SPERMATOZOON BEFORE ACTUAL FERTILIZATION- TAKES PLACE IN FEMALE GENITAL TRACT.
  - INVOLVES NUMBER OF CHANGES:
  - CHANGES IN THE SPERM CELL MEMBRANE
  - SIGNAL TRANSDUCTION (CALCIUM INFLUX, RELEASE OF ACROSOMAL ENZYMES, ETC.).
  - CHANGES IN SURFACE GLYCOPROTEINS BY SECRETIONS OF THE FEMALE GENITAL TRACT THUS ALLOWING SPERM-EGG BINDING.

FOLLICULAR FLUID IS THOUGHT TO ENHANCE THE CAPACITATION.

- OVUM TRANSPORT
  - THE CILIARY BEATS OF FIMBRIAE SWEEP THE OVULATORY MASS INTO THE INFUNDIBULUM.
  - THE CILIARY BEATS OF UTERINE EPITHELIUM AND MUSCULAR CONTRACTIONS OF UTERINE TUBE- RESPONSIBLE FOR TRANSCOELOMIC MIGRATION OF OVUM FROM THE SURFACE OF OVARY INTO THE AMPULLA OF UTERINE TUBE.

CONTACT AND FUSION OF GAMETES: THERE ARE THREE BARRIERS WHICH THE SPERM MUST PENETRATE BEFORE FUSING WITH THE OVUM.

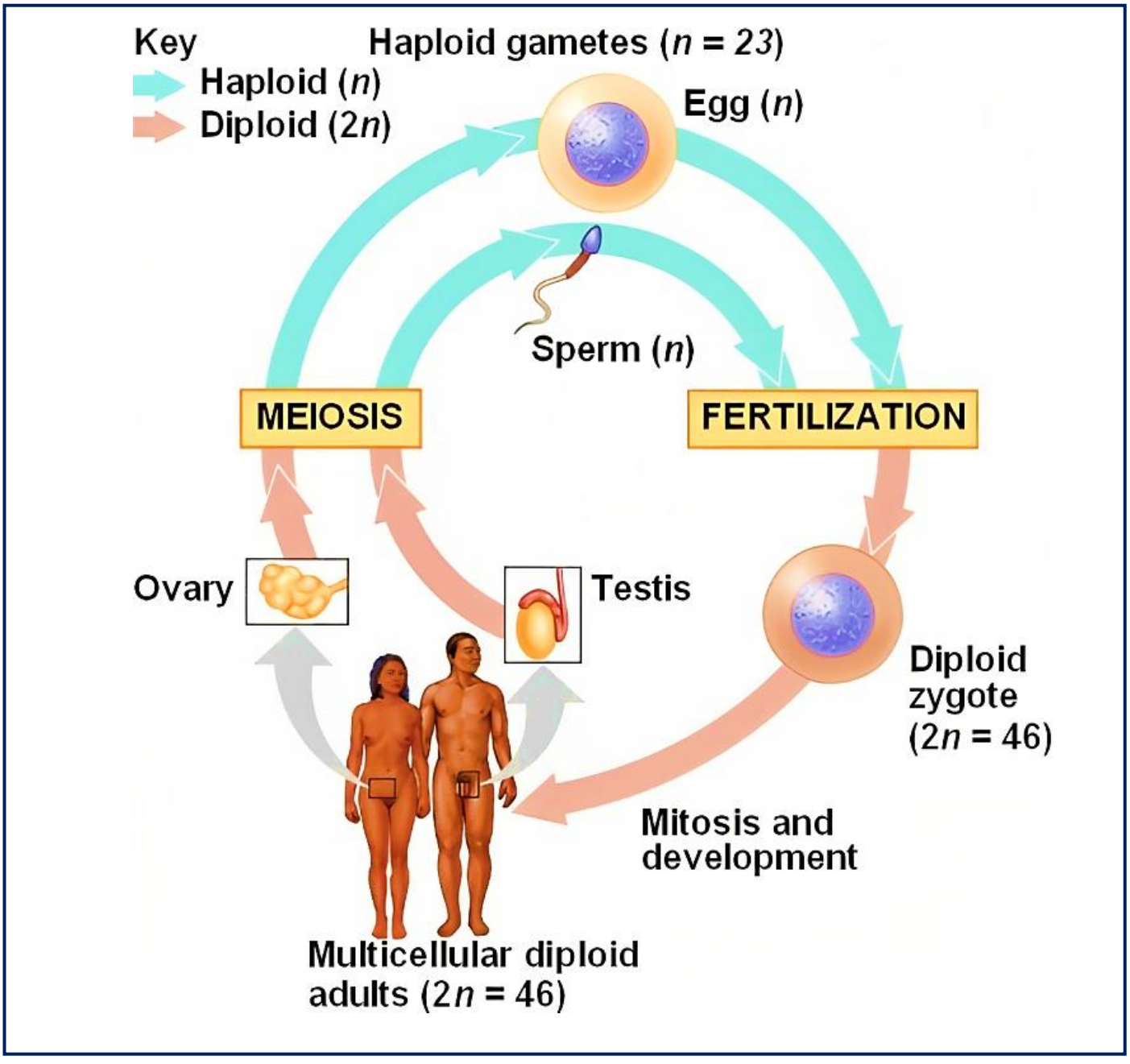
- THEY ARE:
  1. CORONA RADIATA
  2. ZONA PELLUCIDA
  3. VITELLINE MEMBRANE

- FOUR PROCESSES ARE INVOLVED IN THE PENETRATION OF THESE BARRIERS. THEY ARE:
  i. ACROSOME REACTION
  ii. DISINTEGRATION OF BARRIERS
  iii. CALCIUM WAVE IN OOCYTE CYTOPLASM
  iv. NUCLEAR FUSION

**EFFECTS/RESULTS OF FERTILIZATION:** RESULTS OF FERTILIZATION ARE:

i. COMPLETION OF SECOND MEIOTIC DIVISION OF FEMALE GAMETE (SECONDARY OOCYTE)
ii. RESTORATION OF DIPLOID NUMBER (46) OF CHROMOSOMES
iii. DETERMINATION OF CHROMOSOMAL SEX OF THE FUTURE INDIVIDUAL TO BE BORN
iv. INITIATION OF CLEAVAGE (MITOTIC) DIVISION OF ZYGOTE
v. DETERMINATION OF POLARITY AND BILATERAL SYMMETRY OF EMBRYO
vi. GENETIC DIVERSITY.

**EMBRYO CONTAINS ONLY MATERNAL MITOCHONDRIA**



# SPERMATOGENESIS

**PROCESS OF MATURATION OF MALE GAMETES IN THE WALL OF SEMINIFEROUS TUBULES.**

- SERIES OF CHANGES LEADS TO CONVERSION OF SPERMATOGONIA INTO SPERMATOZOA.
- BEGINS- AT THE AGE OF PUBERTY (12-16 YEARS)- CONTINUES EVEN INTO OLD AGE.
- DURATION: 64-74 DAYS.
- FOLLOWING STAGES ARE SEEN
    ii. SPERMATOCYTOSIS
    iii. MEIOSIS
    iv. SPERMIOGENESIS.

- **SPERMATOCYTOSIS:**
  CONVERSION OF SPERMATOGONIA TO PRIMARY SPERMATOCYTES. (BY REPEATED MITOTIC DIVISIONS.
    - PRODUCTION OF TYPE A SPERMATOGONIA MARKS THE BEGINNING OF SPERMATOGENESIS.
    - TYPE A SPERMATOGONIA (44 + X + Y) FORM CLONE OF CELLS BY REPEATED CELL DIVISION.

- **MEIOTIC DIVISIONS:**
  CONVERSION OF PRIMARY SPERMATOCYTES TO SECONDARY SPERMATOCYTES THEN SPERMATIDS.

    - PRIMARY SPERMATOCYTES (44 + X + Y) DIVIDE TO FORM TWO SECONDARY SPERMATOCYTES.
    - FIRST MEIOTIC DIVISION: REDUCES THE NUMBER OF CHROMOSOMES TO HALF.
    - SECONDARY SPERMATOCYTE (22 + X OR 22 + Y) DIVIDES- FORM TWO SPERMATIDS.
    - SECOND MEIOTIC DIVISION: NO REDUCTION IN CHROMOSOME NUMBER.

- **SPERMIOGENESIS:**
  PROCESS OF METAMORPHOSIS (CHANGE IN SHAPE) OF SPERMATIDS TO SPERMATOZOA
    - MAJOR EVENTS IN SPERMIOGENESIS:
        - NUCLEAR MORPHOGENESIS AND CONDENSATION
        - FORMATION OF TAIL- FROM AXIAL FILAMENT
        - FORMATION OF ACROSOME- FROM GOLGI BODY
        - REARRANGEMENT OF ORGANELLES (MITOCHONDRIA, CENTRIOLES)
        - SHEDDING OF EXCESS CYTOPLASM.
        - CENTROSOME: DIVIDES INTO TWO PARTS: PROXIMAL AND DISTAL. THE PROXIMAL CENTRIOLE- SPHERICAL AND LIE IN THE NECK. DISTAL CENTRIOLE FORMS THE DISTAL END OF THE MIDDLE PIECE, I.E., ANNULUS.

CELL MEMBRANE: PERSISTS AS A COVERING FOR THE SPERMATOZOON.

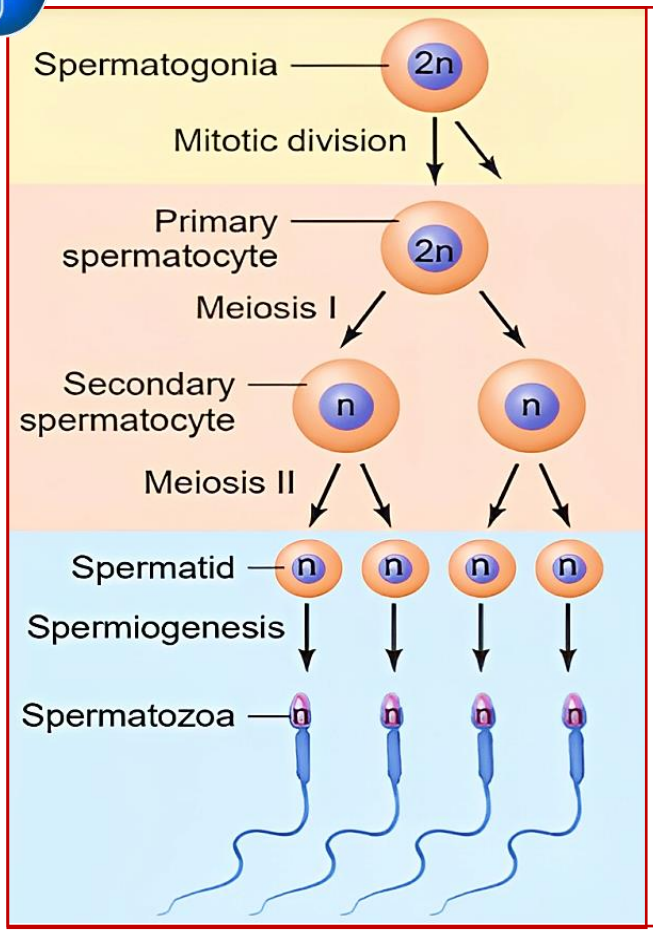
# OOGENESIS

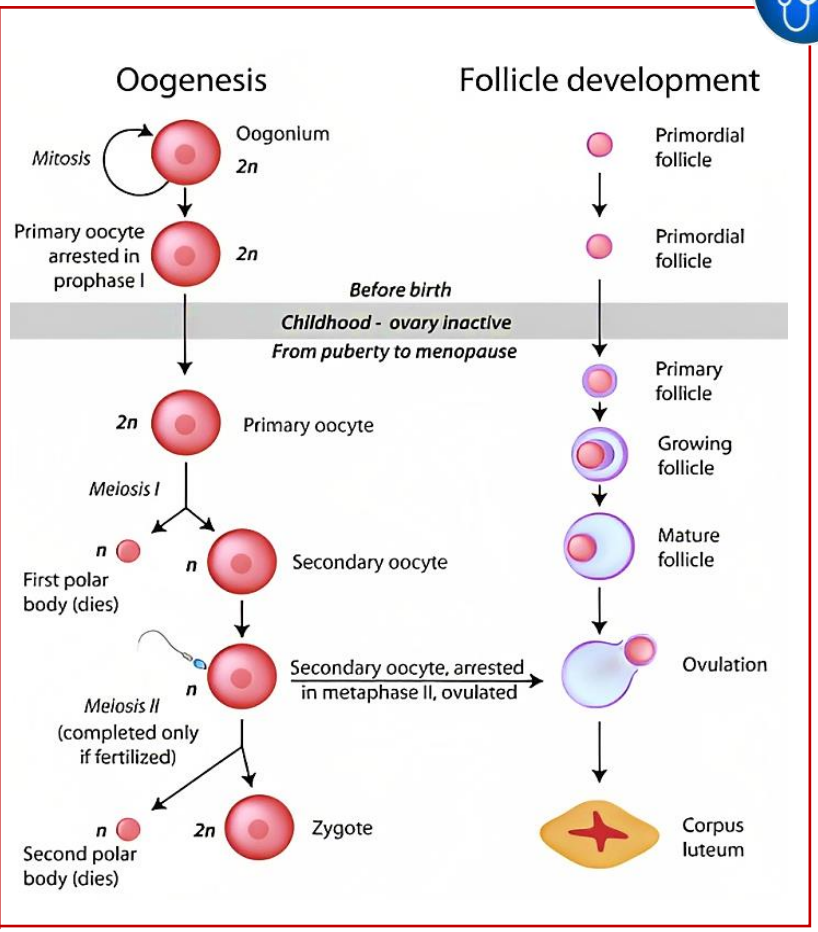
- PROCESS OF MATURATION AND DIFFERENTIATION OF PGC TO OOGONIA, PRIMARY OOCYTES, SECONDARY OOCYTES AND TO MATURE OVA IN THE FEMALE GENITAL TRACT.
    - PECULIARITIES OF OOGENESIS:
        - STARTS BEFORE BIRTH (10TH WEEK)
        - STOPS IN THE MIDDLE (BIRTH TO PUBERTY)
        - RESTARTS AT PUBERTY (11-13 YEARS)
        - CONTINUES UP TO MENOPAUSE (45-55 YEARS)

- PROCESSES: THE VARIOUS PROCESSES IN OOGENESIS ARE:
  - MITOSIS
  - MEIOSIS
  - GROWTH OF FOLLICLES
  - DIFFERENTIATION OF FOLLICLES

- BEFORE BIRTH:
  - BEFORE 3RD MONTH: PGCS UNDERGO MITOSIS- FORM OOGONIA (IN THE ABSENCE OF TESTICULAR DIFFERENTIATION FACTOR (TDF))
  - BEFORE 7TH MONTH: THE OOGONIA CONTINUE TO DIVIDE MITOTICALLY. SOME GET ENLARGED AND FORM PRIMARY OOCYTES.
  - 7TH MONTH TO BIRTH: FORMATION OF PRIMORDIAL FOLLICLES (PRIMARY OOCYTE WITH ITS SURROUNDING FLAT EPITHELIAL CELLS). PRIMARY OOCYTE ENTERS PROPHASE I OF MEIOSIS I- MEIOSIS IS ARRESTED BY OOCYTE MATURATION INHIBITOR (OMI) FACTOR.
  - BIRTH TO PUBERTY: PRIMARY OOCYTES ENTER PROLONGED RESTING (DIPLOTENE) STAGE.

- AFTER PUBERTY—CYCLICAL PREPARATION FOR FERTILIZATION- OVARIAN CYCLE.
  - FROM THE TIME OF BIRTH TO PUBERTY- DEGENERATION OF NUMBER OF PRIMARY OOCYTES.
  - REST OF THE PRIMARY OOCYTES DO NOT COMPLETE THEIR FIRST MEIOTIC DIVISION UNTIL THEY BEGIN TO MATURE AND ARE READY TO OVULATE.
  - THE FIRST MEIOTIC DIVISION OF A PRIMARY OOCYTE PRODUCES TWO UNEQUAL DAUGHTER CELLS WITH HAPLOID CHROMOSOMES (23). THE LARGE CELL, RECEIVING MOST OF THE CYTOPLASM- SECONDARY OOCYTE, AND SMALLER CELL-"FIRST POLAR BODY".
  - SECONDARY OOCYTE IMMEDIATELY ENTERS THE SECOND MEIOTIC CELL DIVISION.
  - REMAINS ARRESTED IN METAPHASE TILL FERTILIZATION OCCURS.
  - THE SECOND MEIOTIC DIVISION IS COMPLETED ONLY IF FERTILIZATION OCCURS.
  - THIS DIVISION RESULTS IN TWO UNEQUAL DAUGHTER CELLS. THE LARGER CELL IS CALLED OVUM. THE SMALLER DAUGHTER CELL IS CALLED THE SECOND POLAR BODY.




# FIRST WEEK OF DEVELOPMENT

**IN THE FIRST WEEK - FERTILIZATION TAKES PLACE IN AMPULLA OF FALLOPIAN TUBE**

**ZONA PELLUCIDA - A MEMBRANE TO PREVENT IMPLANTATION AND POLYSPERMY.**

THE SPERM BINDS TO ZONA PELLUCIDA

↓

TRIGGERS THE RELEASE OF ACROSOMAL ENZYME CALLED – ACROSOMAL REACTION
(ACROSIN, ACID PHOSPHATASE, HYALURONIDASE – ACROSOMAL ENZYME )

↓

THE SPERM PENETRATES THE ZONA PELLUCIDA

↓

THE CORTICAL GRANULES RELEASE ITS SECRETION CALLED - CORTICAL REACTION

↓

THE CORTICAL ENZYMES REACH THE ZONA

↓

ZONA CHANGES PERMEABILITY OF SPERMS CALLED - ZONA REACTION

↓

FUSION OF MALE AND FEMALE PRONUCLEUS

↓

FORMS ZYGOTE

↓

UNDERGOES 1ST CLEAVAGE - 2 CELLS STAGE

↓

AND CLEAVAGE - 4 CELLS STAGE

↓

3RD CLEAVAGE - 8 CELLS STAGE.

## TIME AND SITE OF EVENTS

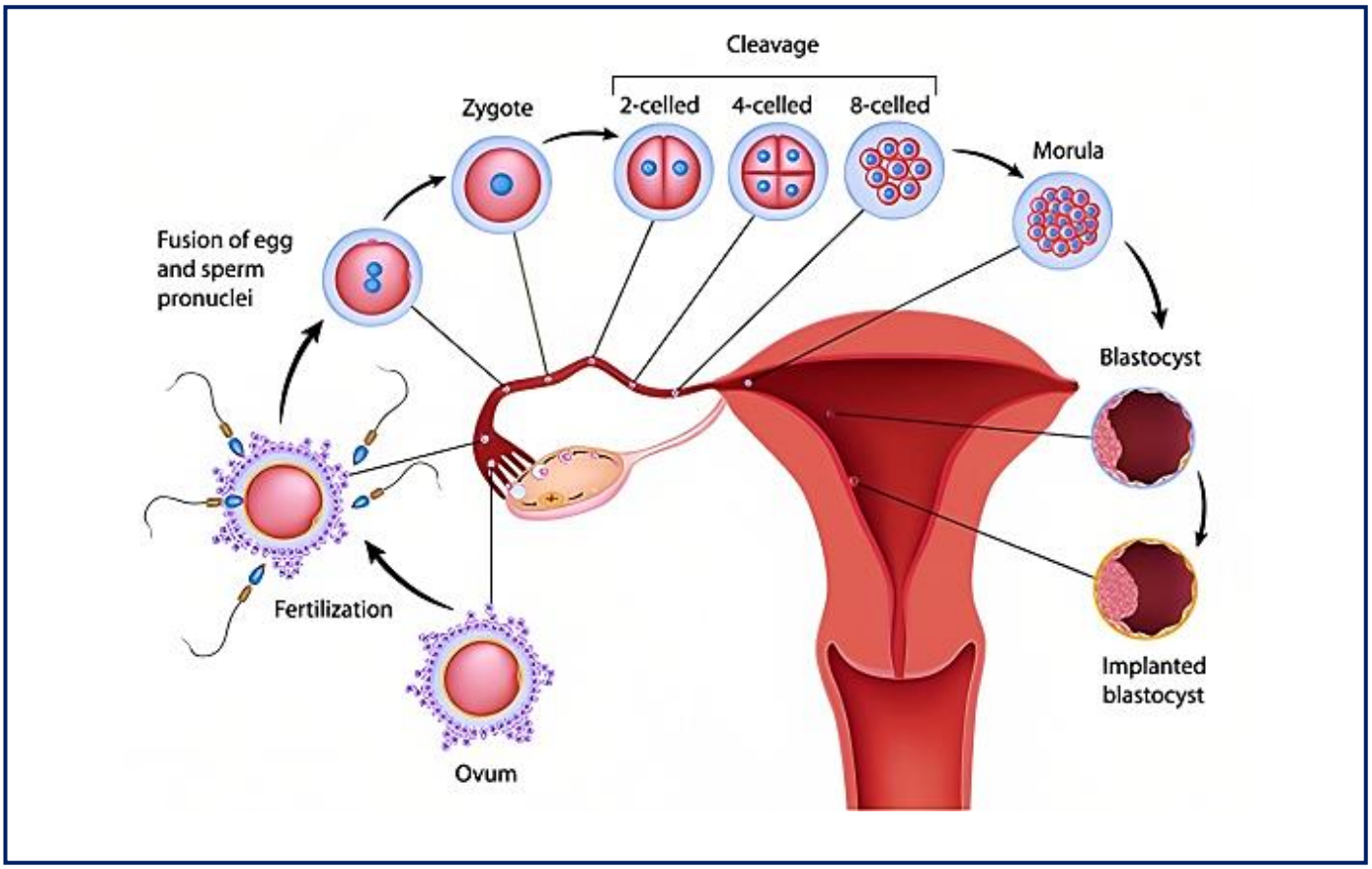
1. FERTILIZATION- TAKES PLACE WITHIN 12 TO 24 HOURS OF OVULATION.
2. ONE DAY AFTER FERTILIZATION- 2 CELL STAGE.
3. THIRD DAY AFTER FERTILIZATION- 8 TO 16 CELLS STAGE [MORULA].
4. FOURTH DAY AFTER FERTILIZATION- 32 CELLS [ADVANCED MORULA] → ENTERS UTERINE CAVITY.

5. UTERINE FLUID ENTERS AND ADVANCED MORULA FORMS BLASTOCYST.

6. AT THE TIME OF FORMATION - BLASTOCYST CONTAINS 58 CELLS:
   53 CELLS IN OUTER CELL MASS FORMS TROPHOBLAST - FORMS PLACENTA.
   5 CELLS IN INNER CELL MASS FORMS EMBRYOBLAST-FORMS EMBRYO PROPER

.

7. BLASTOCYST IS FORMED ON 4TH DAYS AFTER FERTILIZATION.
8. 5TH DAY AFTER FERTILIZATION ZONA PELLUCIDA DISAPPEARS.
9. 6TH TO 7TH AFTER FERTILIZATION IMPLANTATION OCCURS.

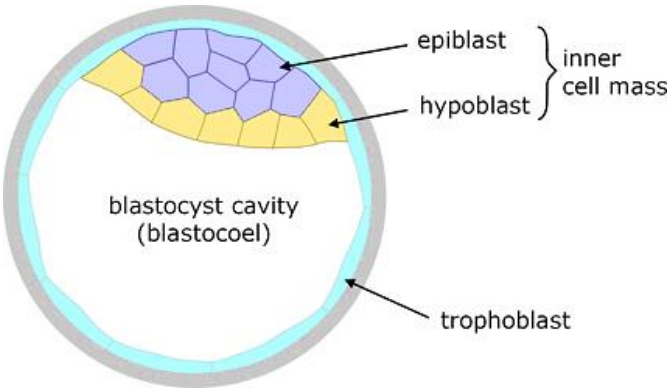



## SECOND WEEK OF DEVELOPMENT

### DEVELOPMENT OF EMBRYOBLAST AND TROPHOBLAST

- THE INNER CELL MASS - FORMS EMBRYOBLAST.
- THE EMBRYOBLAST FORMS TWO LAYERS – COLUMNAR CELLS & CUBOIDAL CELL

1) COLUMNAR CELLS
↓
EPIBLAST CELLS
↓
FORMED ON DORSAL END
↓
ENCLOSES AMNIOTIC CAVITY




2) CUBOIDAL CELLS → HYPOBLAST CELL

HYPOBLAST CELLS
↓
FORMED ON VENTRAL END → FORMS EXOCOELOM MEMBRANE
↓
IT IS FLATTENED CELL → ENCLOSES THE PRIMARY YOLK SAC

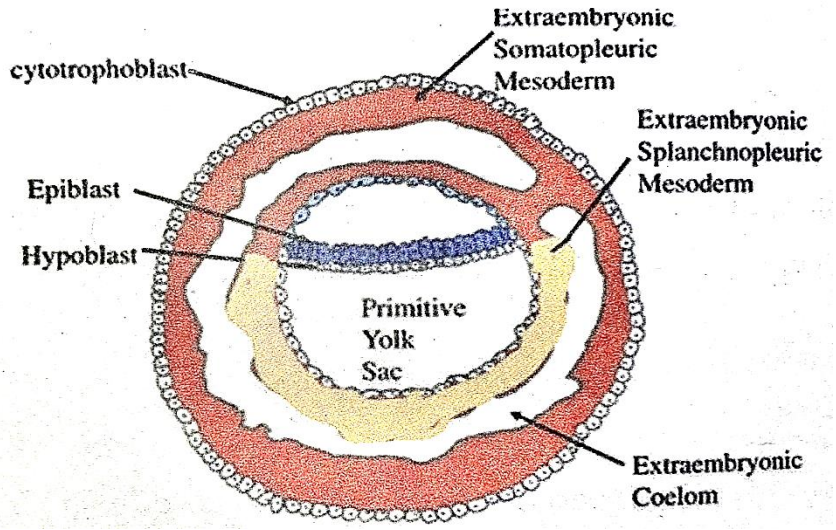
**FORMATION OF SECONDARY YOLK SAC AND UMBILICAL CORD**

- THE SMALL CAVITY FORMED IN THE EEM JOINS TO FORM A LARGER CAVITY.
- THE UNDIVIDED PART OF EEM (EXTRA EMBRYONIC MESODERM) - CRANIAL TO AMNIOTIC CAVITY
  ↓
  FORMS CONNECTING STALK → WHICH BECOMES UMBILICAL CORD.

- THE CAVITY IN EEM DIVIDES IT INTO:
  ↓
  SPLANCHNOPLEURIC LAYER OF EEM- SURROUNDING THE YOLK SAC MAKES THE YOLK SAC SMALLER IN SIZE CALLED THE SECONDARY YOLK SAC.

**SUMMARY OF SECOND WEEK DEVELOPMENT:**

1. TWO LAYERS OF INNER CELL MASS - EPIBLAST
   - HYPOBLAST.
2. TWO LAYERS OF OUTER CELL MASS - CYTOTROPHOBLAST
   - SYNCYTIOTROPHOBLAST.
3. AMNION AND CHORION ARE FORMED.
4. TWO LAYERS OF EEM SOMATIC AND VISCERAL LAYER.
5. YOLK SAC DIVIDED INTO PRIMARY AND SECONDARY YOLK SAC.



# THIRD WEEK OF DEVELOPMENT

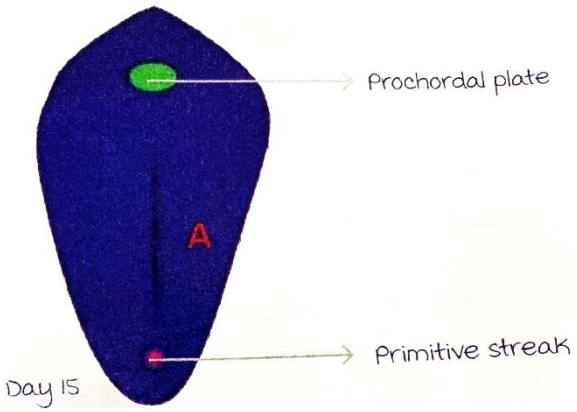
**GASTRULATION & NEURULATION OCCUR IN 3RD WEEK**
- GASTRULATION – FORMATION OF GERM LAYER
- NEURULATION – FORMATION OF NEURAL TUBE

## GASTRULATION

- PRECHORDAL PLATE: COLUMNAR HYPOBLAST CELL
- FROM BUCCOPHARYNGEAL MEMBRANE -> RUPTURES & FORMS ORAL CAVITY

1. HEAD END
 - PRIMITIVE STREAK: PROLIFERATION OF EPIBLAST.
 - PRIMITIVE STREAK INDICATES THE BEGINNING OF GASTRULATION.

2. TAIL END
 - END OF AND WEEK/BEGINNING OF 3RD WEEK




1. 1ST GERM LAYER TO DEVELOP → ENDODERM
2. 2ND GERM LAYER -> MESODERM
 ↑
 DERIVED FROM PRIMITIVE STREAK
 ↑
 DERIVED FROM EPIBLAST

3.LAST GERM LAYER: ECTODERM
 -THE REMAINS OF EPIBLAST AFTER THE FORMATION OF INTRAEMBRYONIC MESODERM.
 - ALL GERM LAYERS DERIVED FROM EPIBLAST > YOLK SAC

## NOTOCHORD DEVELOPMENT

**INTRODUCTION -> MIDLINE STRUCTURE**
- DEVELOP LYING CRANIAL END PRIMITIVE STREAK & CAUDAL END PROCHORDAL PLATE

**STAGE -**

1. CRANIAL END PRIMITIVE STREAK → THICK → CALLED PRIMITIVE KNOT

2. DEPRESSION AREA → CENTER OF PRIMITIVE KNOT CALLED PRIMITIVE PIT

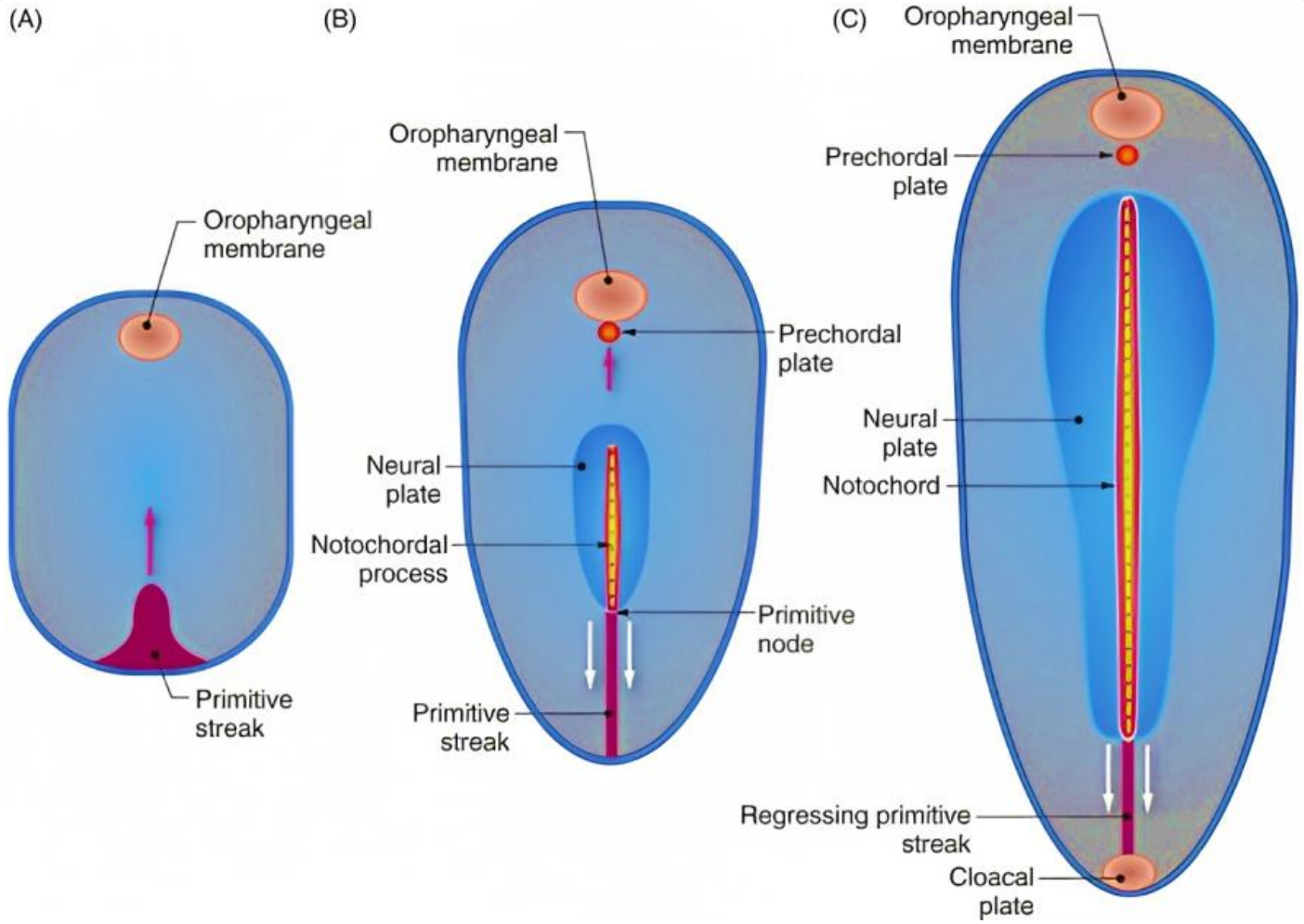
3. CELL OF PRIMITIVE KNOT MULTIPLY → PASSING BETWEEN ECTODERM & ENDODERM TO PRECHORDAL PLATE CALLED NOTOCHORDAL PROCESS

4. CAVITY OF PROMOTIVE POT EXTEND INTO → NOTOCHORDAL PROCESS CALLED NOTOCHORDAL CANAL

5. CELL OF NOTOCHORDAL CANAL → INTERCALATED IN CELL OF ENDODERM AND BREAK DOWN, THIS STAGE FORM COMMUNICATION OF AMNIOTIC CAVITY & YOLK SAC

6. WALL OF CANAL FLATTED → NOTOCHORDAL PLATE

7. CELL PROLIFERATION → FLAT TO CURVE → NOTOCHORD




# PHARYNGEAL ARCHES DERIVATIVES

1.IT IS THE MESENCHYMAL CONDENSATION.
2.THE MESENCHYME IS DERIVED FROM THE NCC, PARAXIAL & LATERAL MESODERM.
3.THERE ARE 6 PHARYNGEAL ARCHES - I, II, III, IV, V, VI.
4.THE 5TH PHARYNGEAL ARCH DISAPPEARS:-
-THE OUTER LAYER - ECTODERM
-THE INNER LAYER – ENDODERM

THE SPACES BETWEEN THE ECTODERM - CLEFT.
THE SPACES BETWEEN THE ENDODERM – POUCHES

## PHARYNGEAL ARCH DERIVATIVES

1. BONE – 1ST → MALLEUS, INCUS, MANDIBLE, MAXILLA, ZYGOMATIC, PALATINE, TEMPORAL
2ND → STAPES, STYLOID, LESSER CORNU OF HYOID, UPPER BODY OF HYOID
3RD → GREATER CORNU OF HYOID & LOWER HALF OF BODY OF HYOID
4TH → THYROID CARTILAGE, ARYTENOID CARTILAGE, CUNEIFORM CARTILAGE

2.MUSCLE- 1ST → MUSCLE OF MASTICATION, TENSOR TYMPANI, TENSOR PLATE, ANTERIOR BELLY OF DIGASTRIC MUSCLE

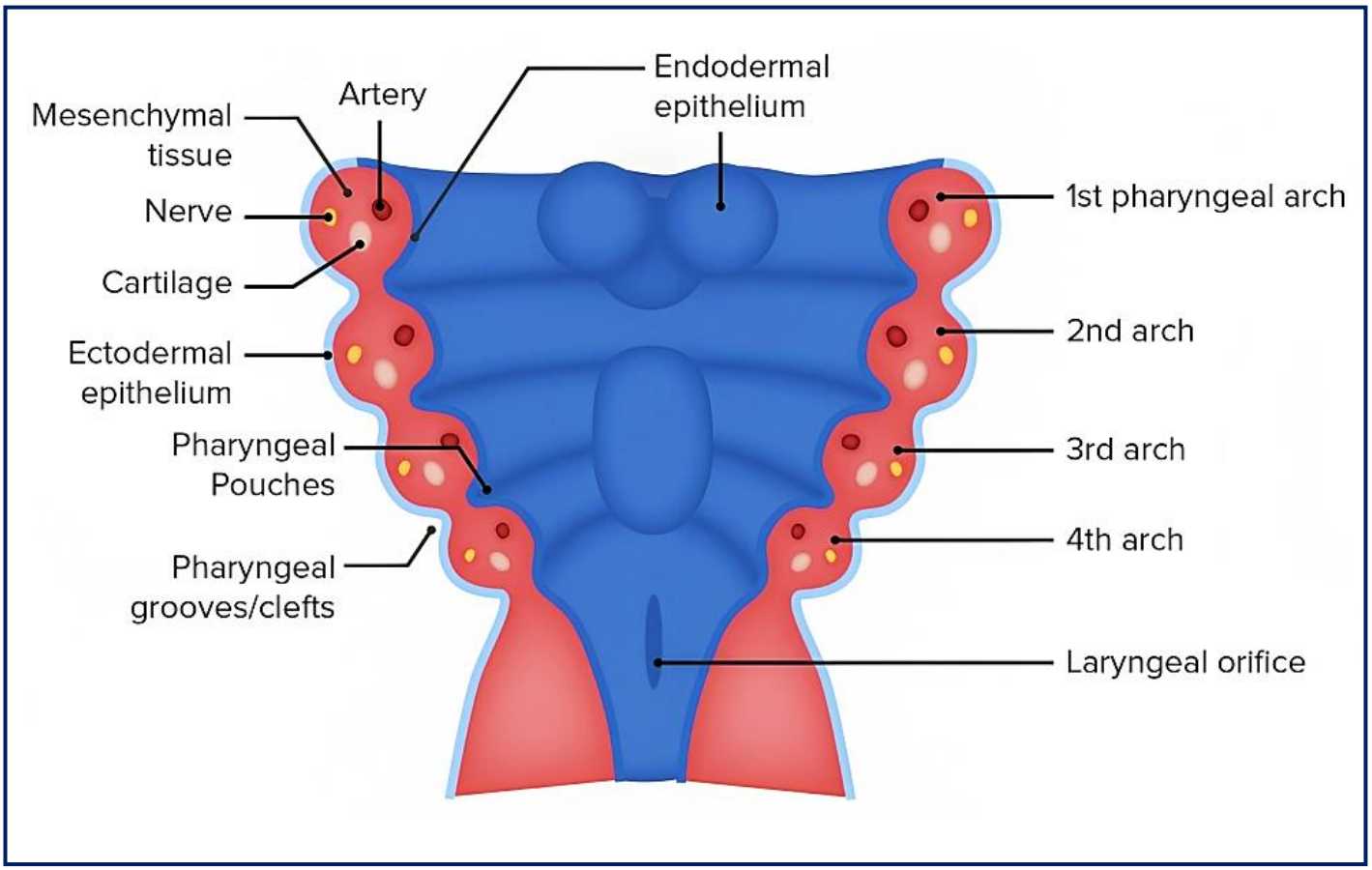
2ND → STAPEDIUS, STYLOHYOID, POSTERIOR BELLY OF DIGASTRIC, FACIAL MUSCLE
3RD → STYLOPHARYNGEUS
4TH → LARYNX MUSCLE

3.NERVE - 1ST → TRIGEMINAL NERVE - MAXILLARY & MANDIBULAR
2ND → FACIAL NERVE
3RD → GLOSSOPHARYNGEAL NERVE
4TH → SUPERIOR LARYNGEAL BRANCH OF VAGUS

4.ARTERY- 1ST → MAXILLARY ARTERY
2ND → HYOID ARTERY, STAPEDIAL ARTERY
3RD → COMMON CAROTID ARTERY & INTERNAL CAROTID ARTERY
4TH → ARCH OF AORTA

## DERIVATIVES OF PHARYNGEAL POUCHES

I POUCH : MIDDLE EAR
EUSTACHIAN TUBE
II POUCH : TONSILLAR FOSSA
TONSILLAR EPITHELIUM
III POUCH : VENTRAL -> THYMUS
DORSAL -> PARATHYROID III BECOMES INFERIOR PARATHYROID
IV POUCH : VENTRAL -> ULTIMOBRANCHIAL BODY PARAFOLLICULAR CELLS
DORSAL -> PARATHYROID IVSUPERIOR PARATHYROID

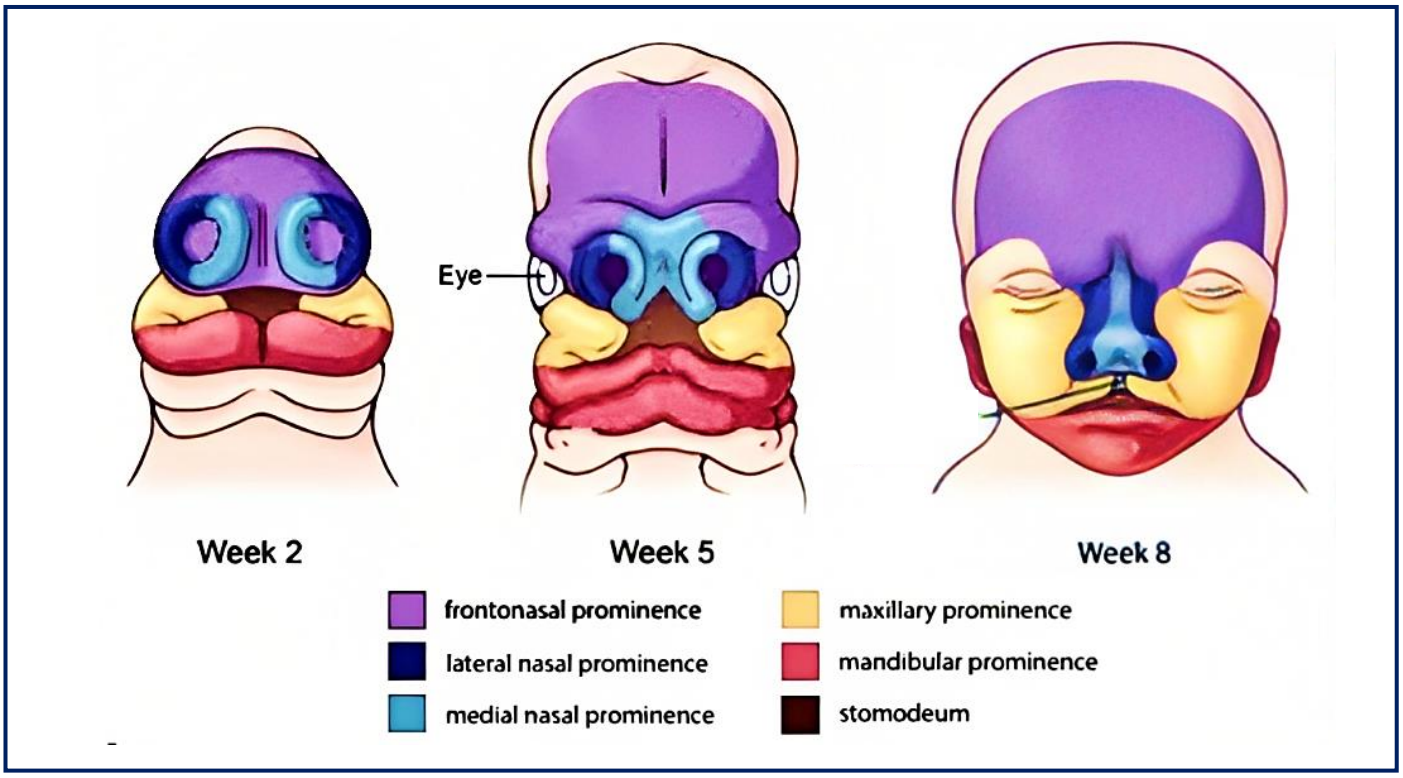


# DEVELOPEMENT OF HEAD AND NECK

## DEVELOPMENT OF FACE

FACE: DERIVED FROM - FRONTONASAL PROCESS
- MEDIAL & LATERAL PROCESS
- MAXILLARY PROCESS
- MANDIBULAR PROCESS
- LATERAL NASAL PROCESS

1. FRONTONASAL PROCESS - FORMS FOREHEAD.
2. MEDIAL NASAL PROCESS - FUSE WITH EACH OTHER
   ↓
   FORMS PHILTRUM AND PRIMITIVE PALATE.
3. LATERAL NASAL PROCESS - FORMS ALA OF THE NOSE.
4. MAXILLARY PROCESS - UPPER LIP AND CHEEK.
5. MANDIBULAR PROCESS - LOWER LIP AND CHIN.
6. THE MAXILLARY PROCESS FUSES WITH MNP AND LNP




## DEVELOPMENT OF SOFT PALATE

PALATE CONSIST OF 2 PART -> PRIMARY PALATE & SECONDARY PALATE
PRIMARY PALATE DEVELOPS FROM → FRONTONASAL PROCESS
SECONDARY PALATE DEVELOPS FROM → MAXILLARY PROCESS

DEVELOPMENT OF PALATE BEGINS IN -> 6TH WEEK
DEVELOPMENT OF PALATE END IN -> 12TH WEEK

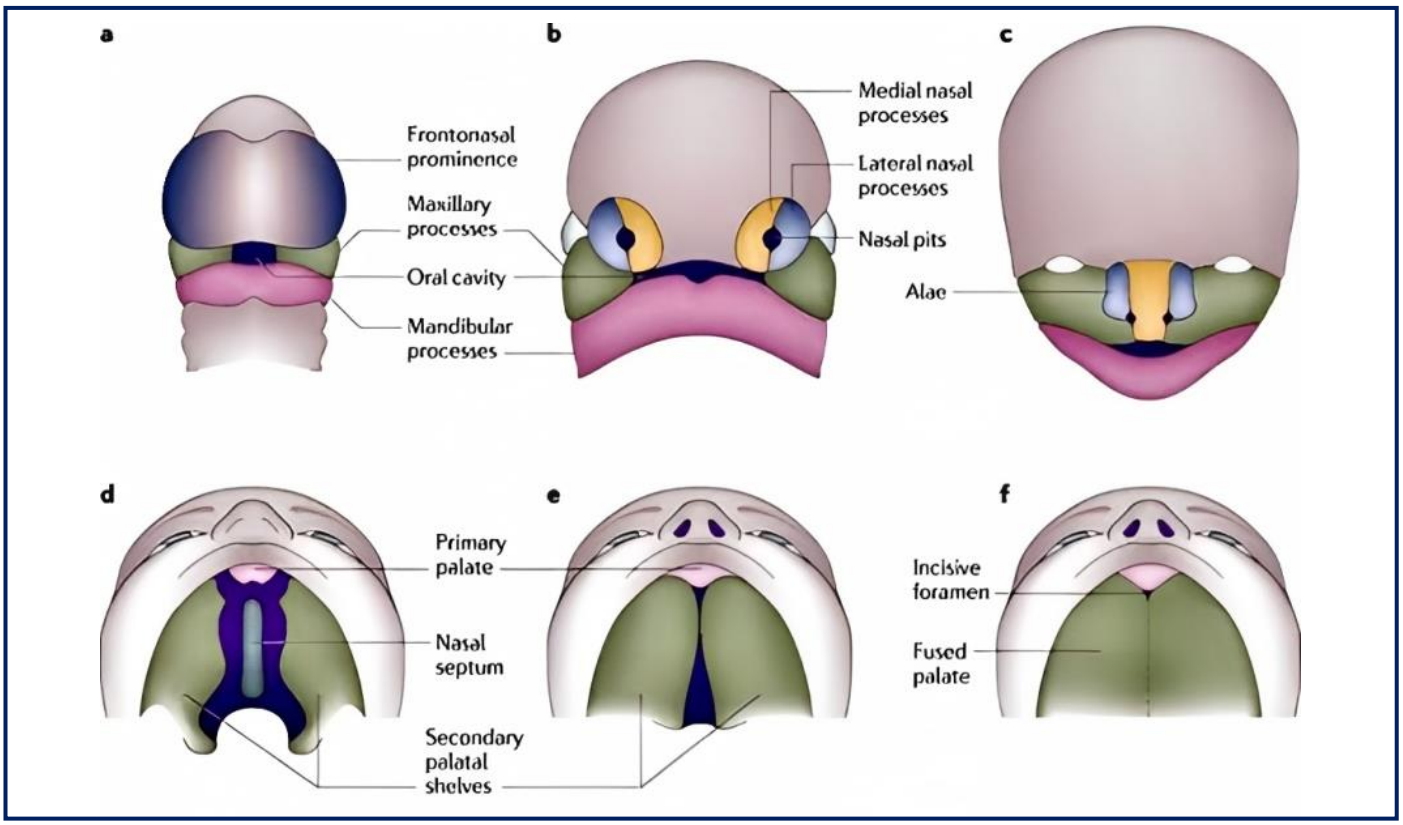
**PALATE DEVELOP IN 2 STAGES-**

1) DEVELOPMENT OF PRIMARY PALATE
2) DEVELOPMENT OF SECONDARY PALATE

DEVELOPMENT OF PRIMARY PALATE -> FORM BY FUSION OF → 2 MEDIAL PROCESS OF FRONTONASAL PROCESS

DEVELOPMENT OF SECONDARY PALATE -> FORM BY FUSION OF 2 SHELVE LIKE OUTGROWTH CALLED PALATINE SHELVES -> APPEAR IN 6TH WEEK

- INTIALLY - GROWTH -> DOWNWARD, MEDIALLY AND BELOW TONGUE
- 7TH AND 8TH WEEK -> ASSUME HORIZONTAL POSITION ABOVE TONGUE & FUSE TO FORM SECONDARY PALATE
- POSTERIOR PART OF THESE → FAIL TO OSSIFY AND FORM -> SOFT PALATE




## CLINICAL ANATOMY



**MIDLINE CLEFT UPPER LIP**
PHILTRUM IS ABSENT
FAILURE OF FUSION OF MNP



**CLEFT UPPER LIP | HARE LIP**
FAILURE OF FUSION OF MAXILLARY PROCESS WITH MNP



**OBLIQUE FACIAL CLEFT**
FAILURE OF FUSION OF MAXILLARY PROCESS WITH LNP

# DEVELOPMENT OF TONGUE

**DEVELOPMENT OF TONGUE**

1) EPITHELIUM OF THE TONGUE - ANTERIOR 2/3RD -> SURFACE ECTODERM.
- POSTERIOR 1/3RD -> ENDODERM.

2) ALL THE MUSCLES ARE DERIVED FROM MYOTOME OF OCCIPITAL SOMITES [EXCEPT PALATOGLOSSUS (IV PHARYNGEAL ARCH].

3) NERVE SUPPLY OF THE TONGUE
- ANTERIOR 2/3RD TONGUE : MANDIBULAR NERVE-LINGUAL NERVE.
- POSTERIOR 1/3RD TONGUE : 9TH NERVE/GLOSSOPHARYNGEAL NERVE.
- POSTERIOR MOST PART OF TONGUE:10TH NERVE.

ON THE FLOOR OF IST ARCH THERE ARE THREE SWELLINGS -

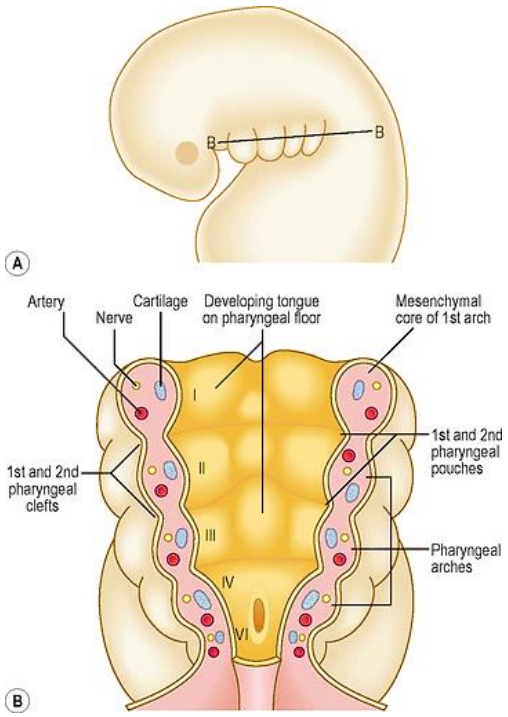
1.TWO LINGUAL SWELLINGS

2.ONE TUBERCULUM IMPAR
↓
FORMS ANTERIOR 2/3RD OF TONGUE

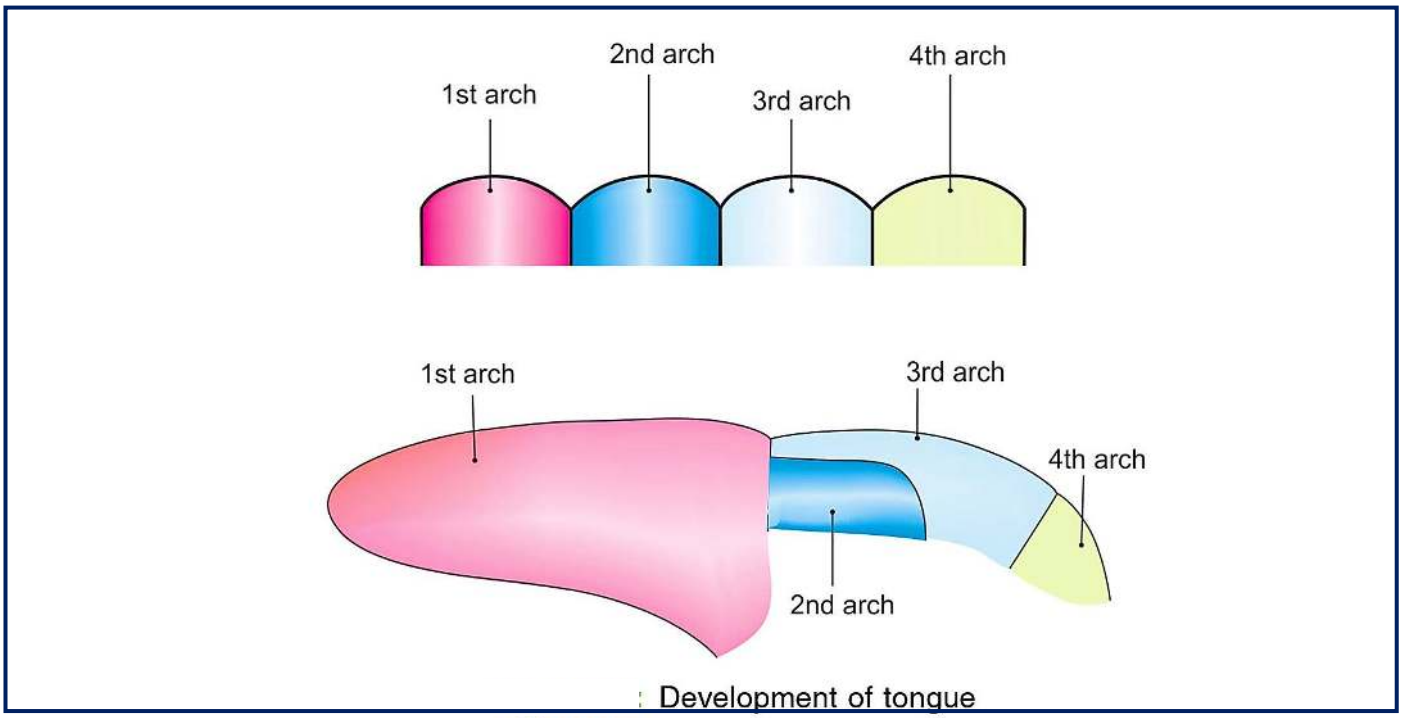
3.FLOOR OF II, III, IV ARCHES - HYPOBRANCHIAL EMINENCE

4.THE III ARCH MESODERM OF HYPOBRANCHIAL EMINENCE FORMS
↓
POSTERIOR 1/3RD OF THE TONGUE.

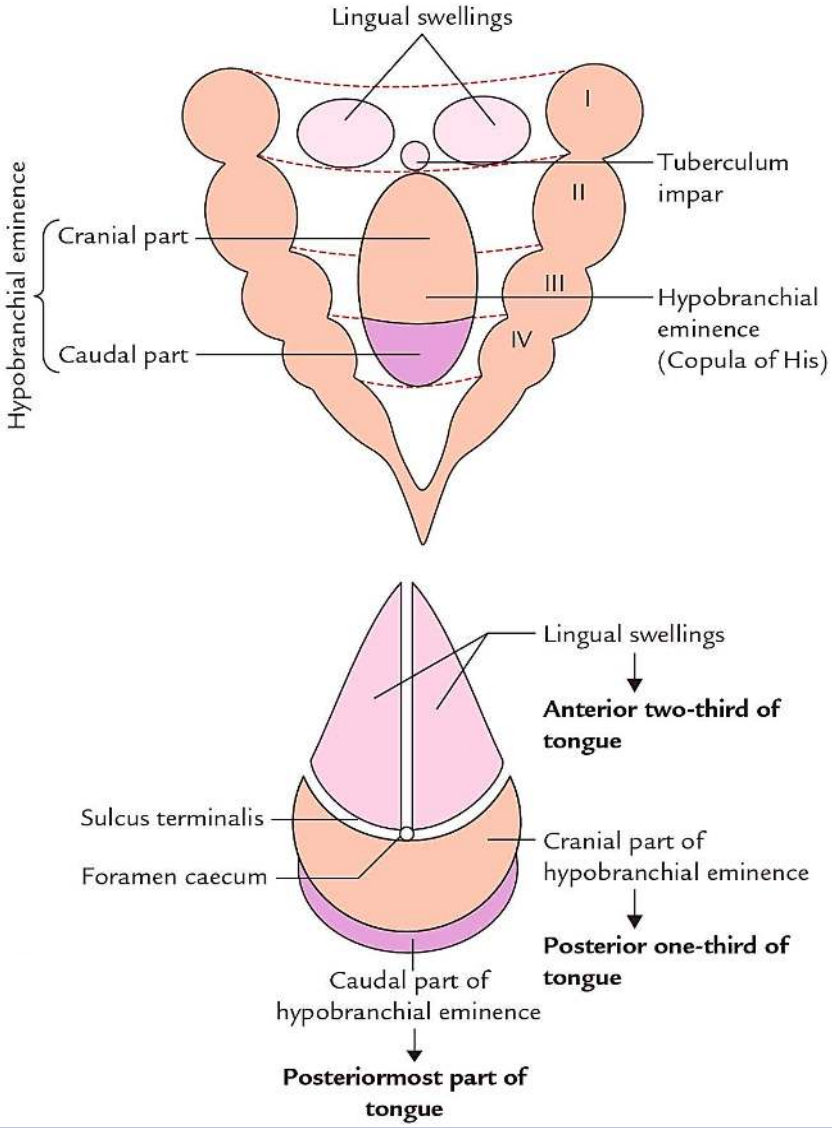
5.THE IV ARCH MESODERM OF HYPOBRANCHIAL EMINENCE FORMS
↓
POSTERIOR MOST PART OF THE TONGUE AND EPIGLOTTIS.




6.THE II ARCH MESODERM DOES NOT CONTRIBUTE TO TONGUE DEVELOPMENT
↓
BECAUSE MESODERM OF III ARCH OVERGROWS II & FUSE WITH ANTERIOR 1/3RD OF TONGUE



Development of tongue




## DEVELOPMENT OF THYROID

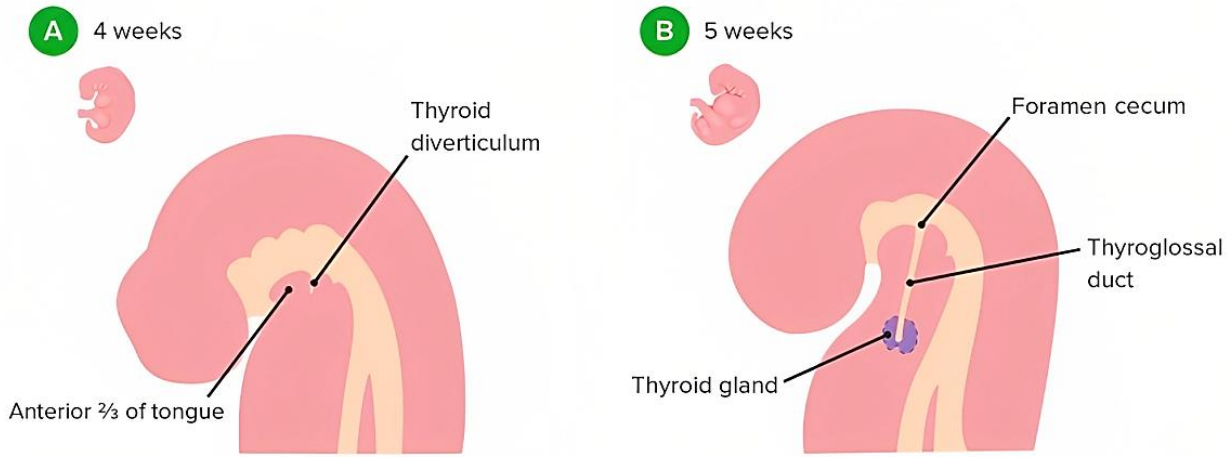
1.THYROID GLAND → DEVELOP FROM → THYROGLOSSAL DUCT
2.PARAFOLLICULAR CELL → DERIVED FROM → CAUDAL PHARYNGEAL COMPLEX
3.MIDLINE OF MANDIBULAR ACHES → SEPARATED BY MIDLINE SWELLING → TUBERCULUM IMPAR → BEHIND DEPRESSED SURFACE→ TO FORM THYROGLOSSAL DUCT

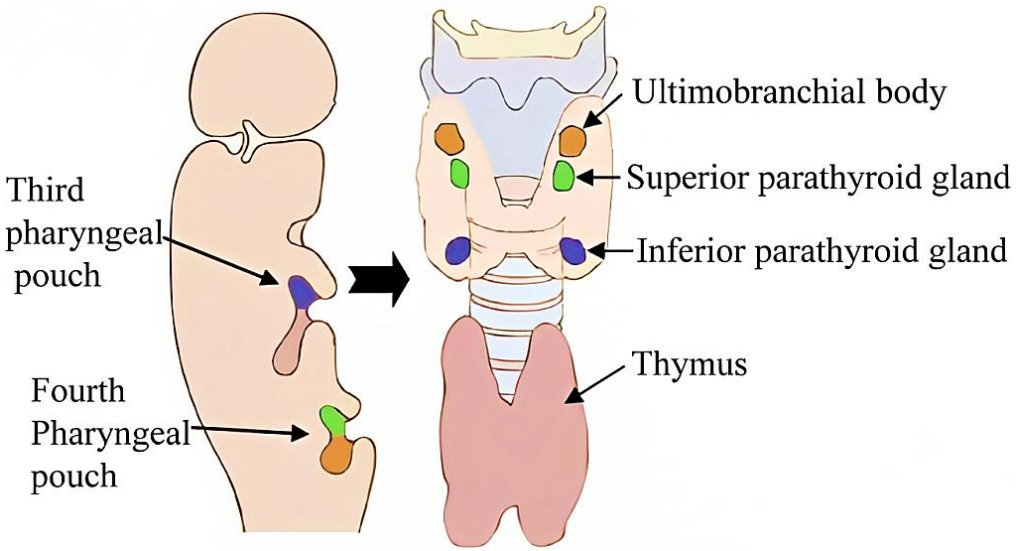
4.SITE OF ORIGIN OF DIVERTICULUM → CALLED FORAMEN CAECUM
5. DIVERTICULUM GROW DOWN → MIDLINE INTO NECK

6. PROLIFERATION OF DIVERTICULUM CELL → FORM 2 LOBES OF THYROID GLAND
7. THYROID COME → INTIMATE RELATIONSHIP WITH CAUDAL PHARYNGEAL COMPLEX → FUSE TO FORM→ PARAFOLLICULAR CELL

## DEVELOPMENT OF PARATHYROID

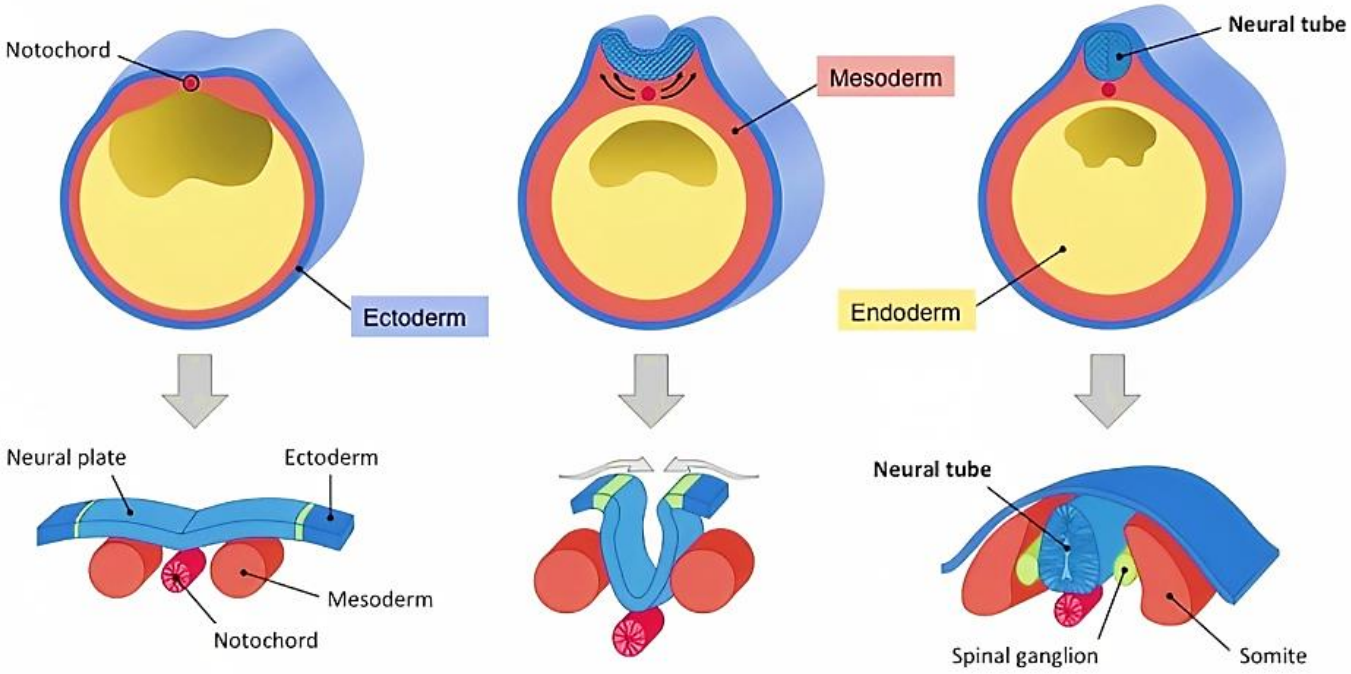
EMBRYOLOGICALLY, THE PARATHYROID GLANDS DERIVE FROM THE ENDODERM OF THE THIRD AND FOURTH PHARYNGEAL POUCHES.
THE THIRD PHARYNGEAL POUCH GIVES RISE TO THE INFERIOR PARATHYROID GLANDS, WHILE THE SUPERIOR PARATHYROIDS ARISE FROM THE FOURTH PHARYNGEAL POUCH




# DEVELOPEMENT OF CNS

## NEURULATION




**1.** Notochord forms from mesoderm cells soon after gastrulation is complete

**2.** Signals from notochord cause inward folding of ectoderm at the neural plate

**3.** Ends of neural plate fuse and disconnect to form an autonomous neural tube

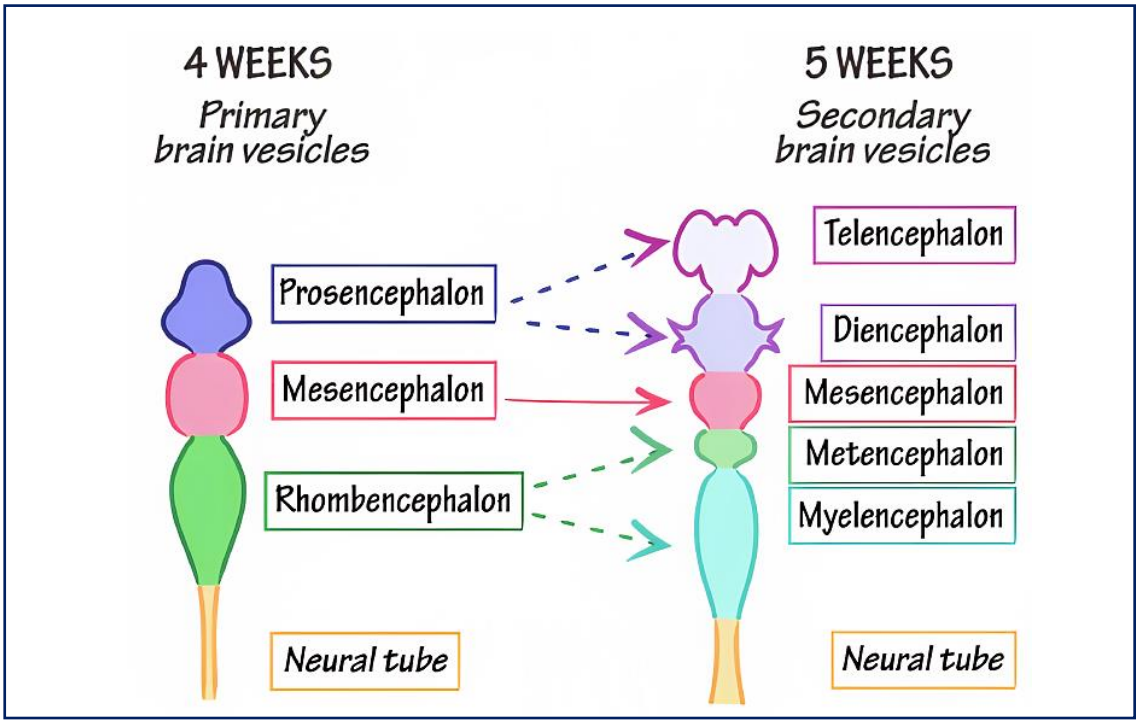
TAKES PLACE AT 3RD WEEK ( ALONG WITH GASTRULATION ) -> FORMATION OF NEURAL TUBE

NOTOCHORD GIVES SIGNAL TO ECTODERM

ECTODERM DIVIDES INTO SURFACE AND NEURO ECTODERM
NEURAL PLATE WILL INVAGINATE DUE TO SINGNAL FROM THE NOTOCHORD -> NEURAL GROOVE

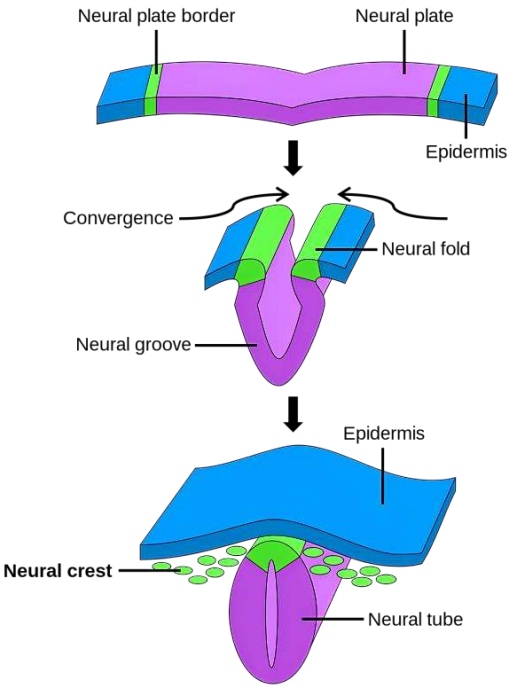
ECTODERM FORMS THE NEURAL PLATE → NEURAL TUBE
NEURAL TUBE -> DIVIDE INTO 2 PARTS -> CRANIAL PART & CAUDAL PART
CAUDAL PART -> FORM SPINAL CORD

CRANIAL PART -> FOR BRAIN
-> CRAINAL PART OF NEURAL TUBE WILL FORM SOME VESICLES HERE




## NEURAL CREST DERIVATIVE

1) NEURON OUT SIDE CNS
2) DORSAL ROOT OF GANGLION
3) SENSORY GANGLION OF CRANIAL NERVE 5TH , 7TH ,10TH
4) SYMPATHETIC & PARASYMPATHETIC NEURON AND GANGLION
5) NEUROGLIA OF PNS → SATELLITE CELL



# DEVELOPEMENT OF CVS

## HEART TUBE FORMATION

1. BULBOUS CORDIS
   - A. PROXIMAL 1/3 WILL FORM THE ROUGH PART OF THE RIGHT VENTRICLE
   - B. CONUS PART WILL FORM INFUNDIBULUM OF BOTH RIGHT VENTRICLE AND LEFT VENTRICLE
   - C. SPIRAL SEPTUM [DERIVED FROM NEURAL CREST CELL] IS DEVELOPED IN TRUNCUS ARTERIOSUS AND DIVIDES IT INTO AORTA AND PULMONARY TRUNK

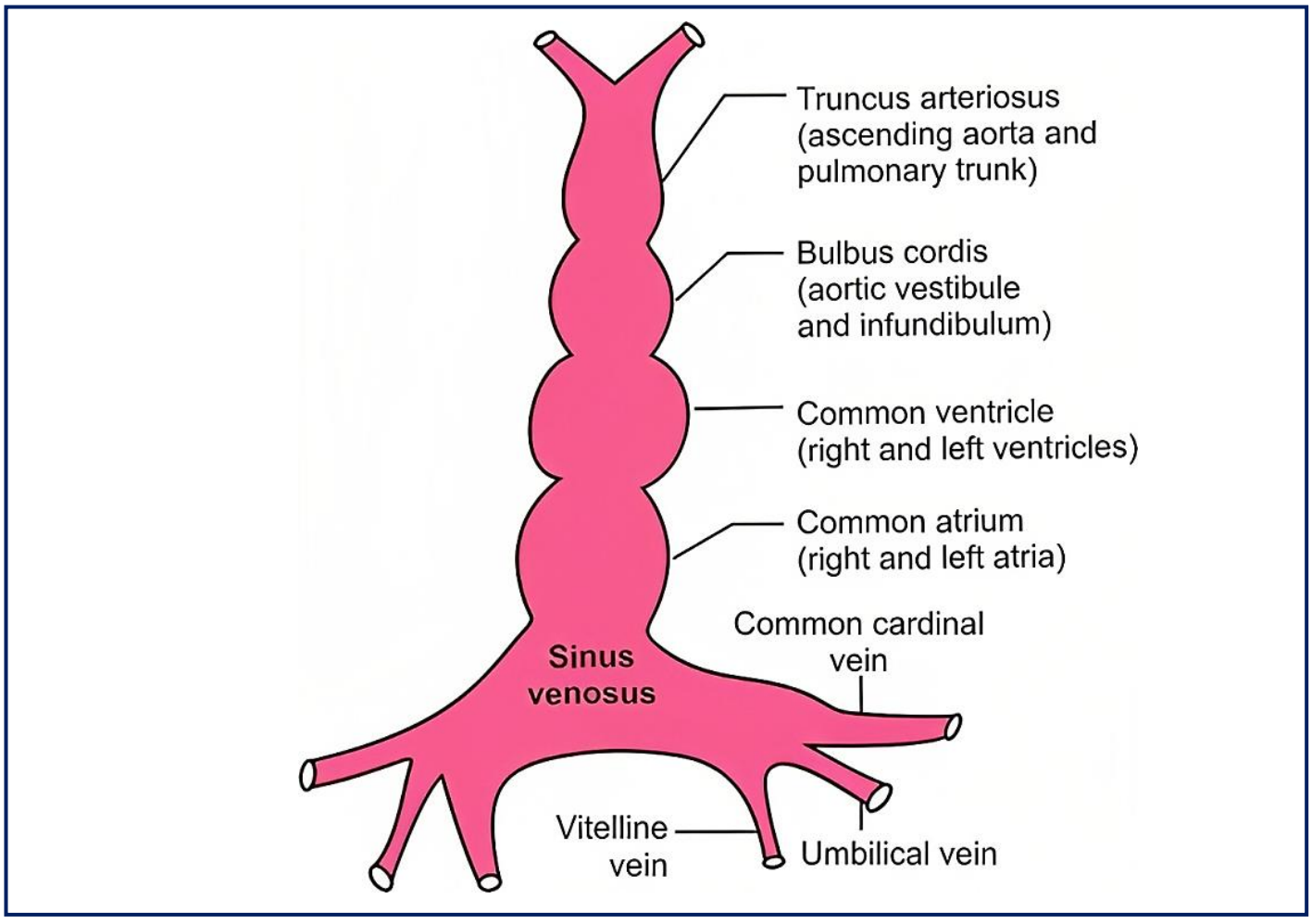
2. PRIMITIVE VENTRICLE: DEVELOPS INTO ROUGH PART OF LEFT VENTRICLE

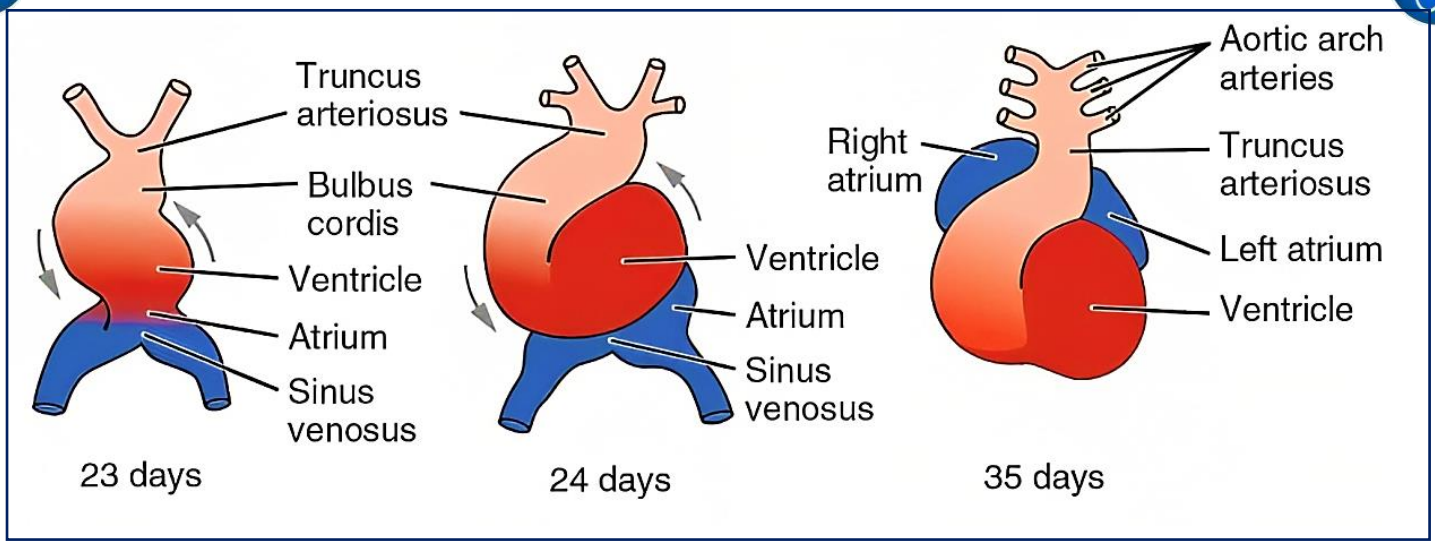
3. BULBO VENTRICULAR SULCUS: FILLED WITH SEPTUM WHICH FORMS INTERVENTRICULAR SEPTUM

4. PRIMITIVE ATRIUM: WILL FORM THE ROUGH PART OF BOTH RIGHT AND LEFT ATRIUM

5. SINUS VENOSUS:
   - A. BODY ABSORBED INTO RIGHT ATRIUM ALONG WITH RIGHT HORN
   - B. LEFT HORN OBLITERATES AND FORMS CORONARY SINUS
   - C. RIGHT COMMON CARDINAL VEIN WILL FORM SUPERIOR VENA CAVA
   - D. RIGHT VITELLINE VEIN WILL FORM INFERIOR VENA CAVA (CRANIAL PART)

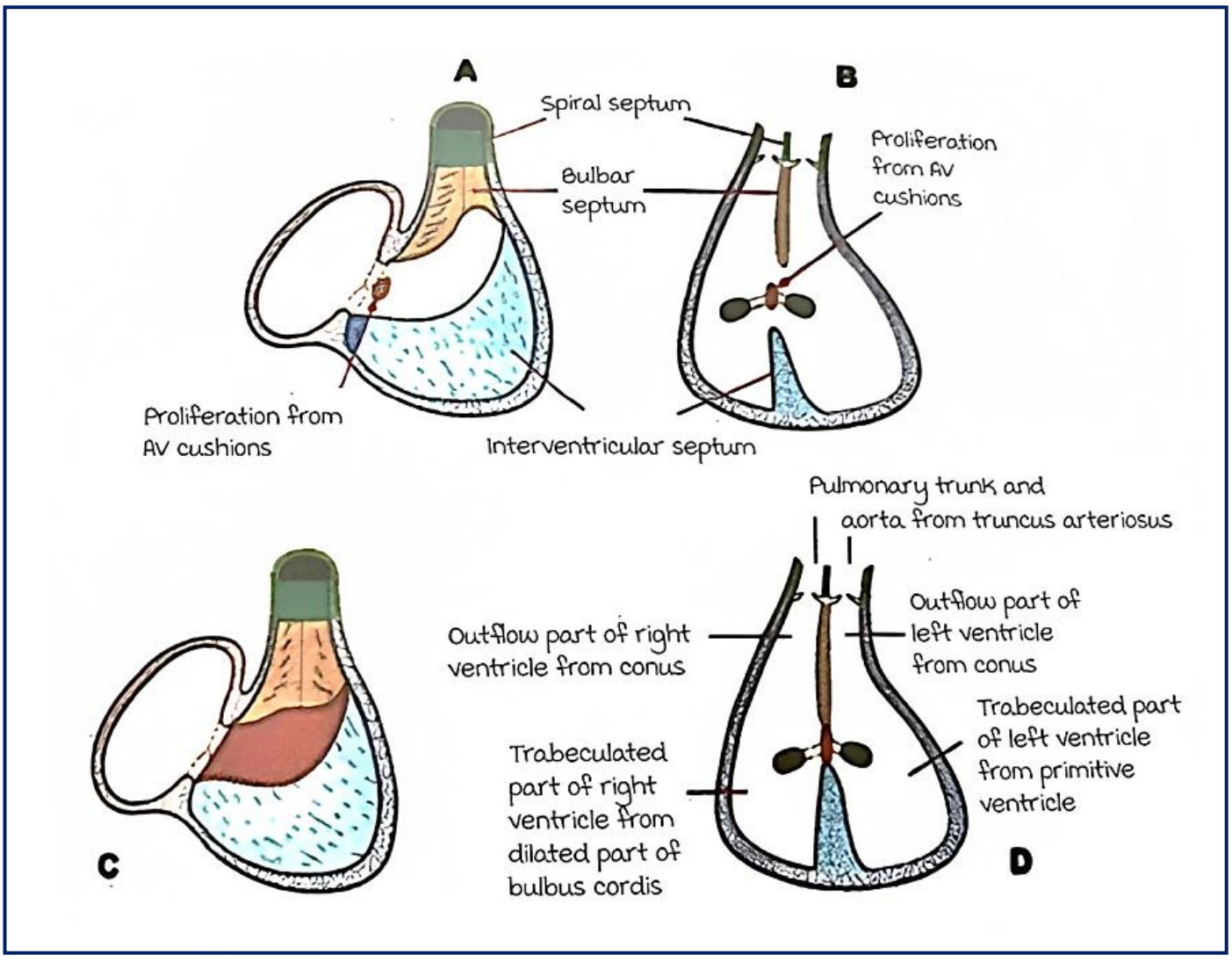
*LOOPING OF HEART – DUE TO DEXTRO LOOPING ATRIAL CHAMBER GOES ABOVE AND BEHIND, VENTRICLE CHAMBER MOVE FRONT AND BELOW






# INTERVENTRICULAR SEPTUM FORMATION

1) **MUSCULAR PART: FROM THE MUSCULAR PROLIFERATION OF BULB VENTRICULAR CAVITY**
2) **MEMBRANOUS PART: DERIVED FROM PROLIFERATION OF AV CUSHION**
3) **BULBAR PART: DERIVED FROM SPIRAL SEPTUM**
   - **PROXIMAL 1/3 OF BULBOUS CORDIS (RV) + PRIMITIVE VENTRICLE (LV)**
     ↓
     **COMMON VENTRICULAR CHAMBER**



# RIGHT ATRIUM DEVELOPMENT

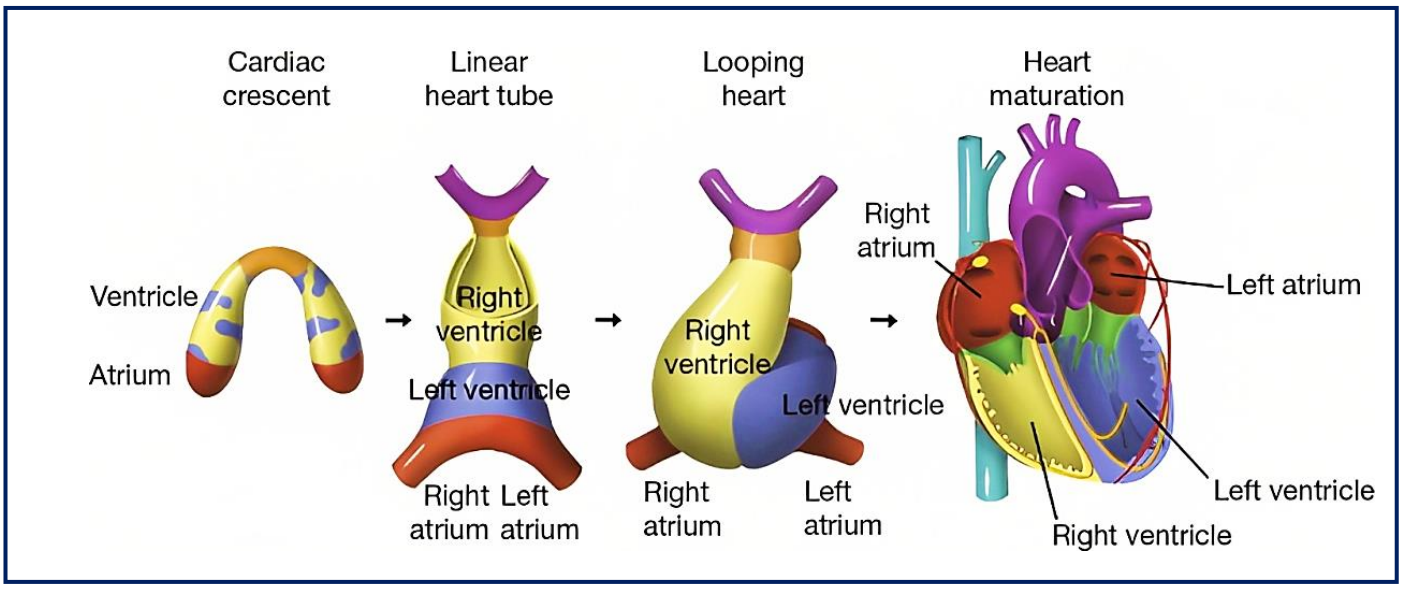
RIGHT ATRIUM → 3 SOURCE

1. RIGHT HALF OF PRIMITIVE ATRIUM - FROM ROUGH TRABECULATED PART
   - IN FRONT CRISTA TERMINALIS
2. ABORPTION OF RIGHT HORN OF SINUS VENOSUS -> SMOOTH PART BEHIND CRISTA TERMINALIS
3. ABORPTION OF RIGHT HALF OF AV CANAL FORM-> MOST VENTRAL → SMOOTH PART

ABSORPTION OF SINUS VENOUS -> CORONARY SINUS & VENA CAVA ARE SEEN

RIGHT VENOUS VALVE → BECOME -> CRISTA TERMINALIS, INFERIOR VENA CAVA

LEFT VENOUS VALVE → FUSED WITH ARTEAL SEPTUM




# DEVELOPMENT OF FOREGUT

FOREGUT TUBE -> RESPIRATORY DIVERTICULUM
-> HEPATIC BUD
-> VENTRAL AND DORSAL PANCREATIC BUD

# DEVELOPMENT OF STOMACH

ROTATES 2 TIMES IN CLOCKWISE DIRECTION

1ST ROTAION -> VERTICAL AXIS

ANTERIOR BORDER BECOMES RIGHT BORDER
POSTERIOR BORDER BECOMES LEFT BORDER
LEFT BORDER -> GROWS RAPIDLY , BECOMES SUPERFICIAL
RIGHT BORDER -> GROWS SLOWLY , MORE POSTERIOR
LEFT VAGUS -> SUPER FICIAL - ANTEROIR VAGAL TRUNK
RIGHT VAGUS -> POSTERIOR VAGAL TRUNK

2ND ROTATION - ANTERO - POSTERIOR AXIS

UPPER END ( CARDIAC END ) BECOMES LEFT SIDE
LOWER END ( PYLOURS ) BECOMES RIGHT SIDE

# DEVELOPMENT OF PANCREAS

PANCREAS DEVELOPMENT FROM - TWO BUDS -> DORSAL PANCREATIC BUD
-> VENTRAL PANCREATIC BUD

VENTRAL BUD MOVES DORSALLY, LIES BELOW THE DORSAL BUD IN -> 7TH WEEK

1.VENTRAL BUD - PART OF HEAD OF PANCREAS + UNCINATE PROCESS

2.DORSAL BUD- REMAINING ENTIRE PANCREAS

PANCREATIC DUCT

1.MAIN PANCREATIC DUCT DERIVED FROM -> DISTAL PART OF DORSAL BUD & WHOLE OF VENTRAL BUD

2.ACCESSORY PANCREATIC DUCT -> DERIVED FROM - PROXIMAL PART OF DORSAL BUD

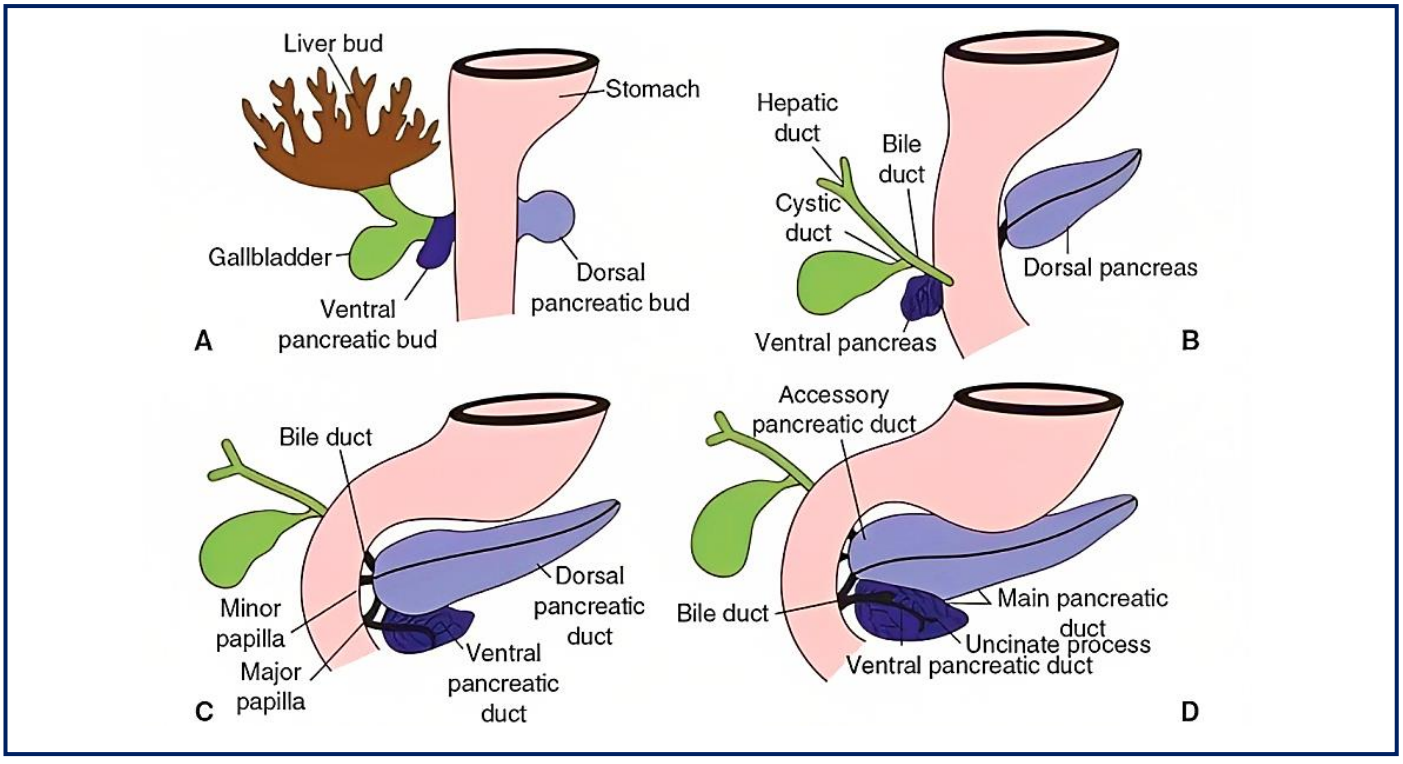
PARENCHYMA DEVELOPS -> FROM -> SURROUNDING MESODERM

# DEVELOPMENT OF LIVER

DEVELOPMENT OF LIVER, FROM A SOURCES - LIVER BUD AND SEPTUM TRANSVERSUM

1.LIVER BUD -> IN FOREGUT TUBE, FORMS PARENCHYMA OF LIVER & BILIARY SYSTEM

2.SEPTUM TRANSVERSUM -> FORMS CONNECTIVE TISSUE OF LIVER, CAPSULE, SINUSOIDS, KUPFFER CELLS.




# DEVELOPMENT OF DUODENUM

DUODENUM → DERIVED FROM FOREGUT + MIDGUT

DUODENUM SUPPLY -> COELIAC TRUNK & SUPERIOR MESENTERIC ARTERY

DURING ROTATION OF STOMACH, DUODENUM COMES TO RIGHT SIDE, FORMS A C - SHAPED LOOP

MESODUODENUM SUSPENDS DUODENUM FROM POSTERIOR ABDOMINAL WALL.

↓

DISAPPEARS- DUODENUM BECOMES RETROPERITONEAL

DUODENUM RETROPERITONEAL EXCEPT THE 1ST PROXIMAL PART OF DUODENUM

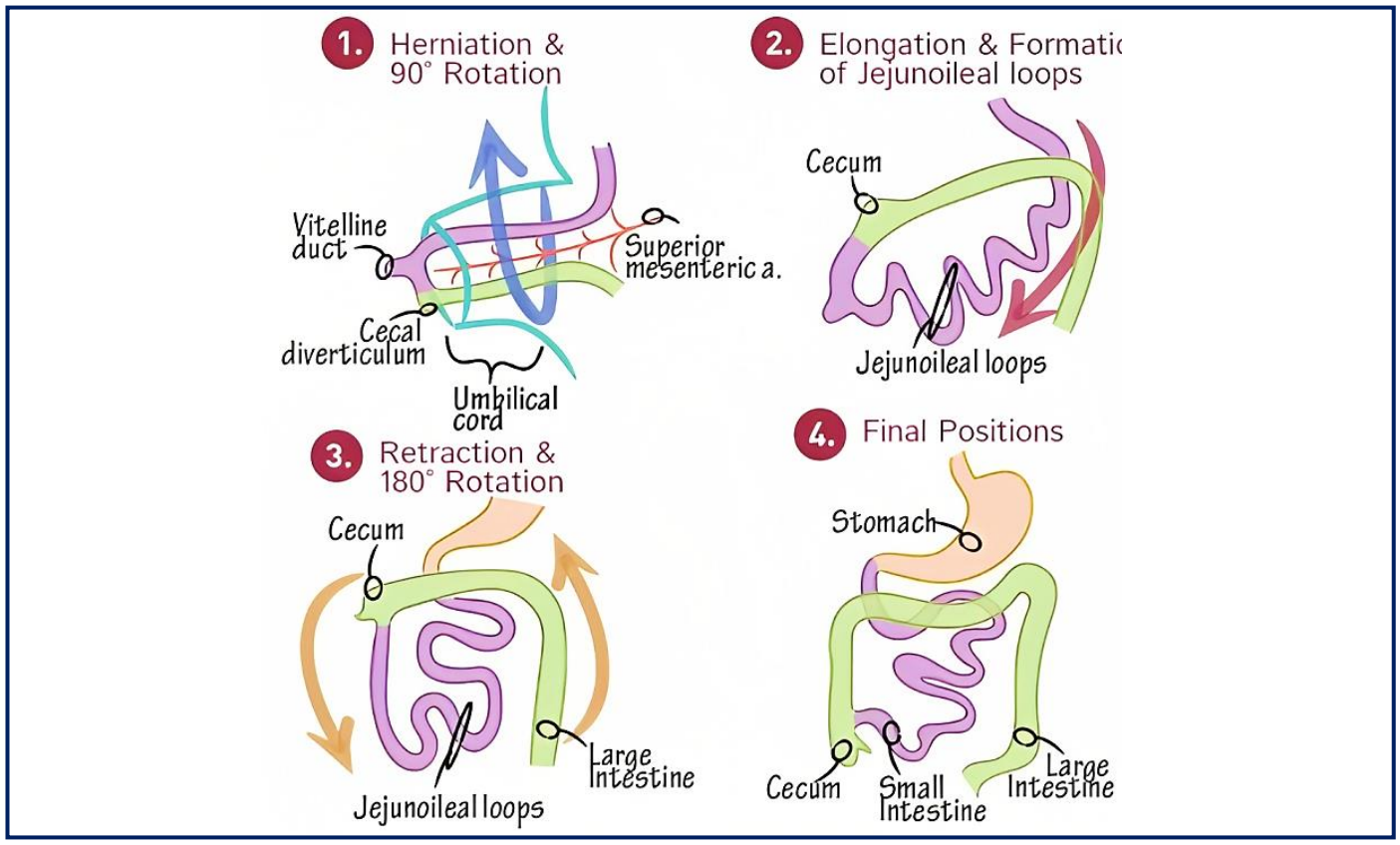
## ROTATION OF MIDGUT

DURING REDUCTION, MIDGUT ROTATES 3 TIMES, EACH TIME BY 90° IN COUNTERCLOCKWISE/ANTI-CLOCKWISE MANNER.

1) 1ST ROTATION -> PRE ARTERIAL SEGMENT -> RIGHT SIDE
-> POST ARTERIAL SEGMENT -> LEFT SIDE

2) 2ND ROTATION -> PRE ARTERIAL SEGMENT - ELONGATES AND ENTERS ABDOMINAL CAVITY AND ROTATES BY 90° ANTI-CLOCKWISE.
-> PRE ARTERIAL SEGMENT COMES TO LIE BEHIND THE SUPERIOR MESENTERIC ARTERY.

3.3RD ROTATION: -POST ARTERIAL SEGMENT -> ENTERS ABDOMINAL CAVITY AND ROTATES BY 90° ANTI- CLOCKWISE.




## DEVELOPMENT OF HINDGUT

## DEVELOPMENT OF KIDNEY

NEPHROGENIC CORD – FORMS → KIDNEY, MESONEPHRIC DUCT & PARAMESONEPHRIC DUCT.

1.AT 4TH WEEK -> PRONEPHROS APPEARS, FUNCTIONS AS THE KIDNEY & DISAPPEARS BY END OF 4TH WEEK.

2.BY THE END OF 4TH WEEK -> MESONEPHROS APPEARS, FUNCTIONS AS INTERIM KIDNEY UP TO 2 MONTHS.

3.BY 5TH WEEK -> METANEPHROS APPEAR AND STARTS ITS FUNCTION BY 3RD MONTH.

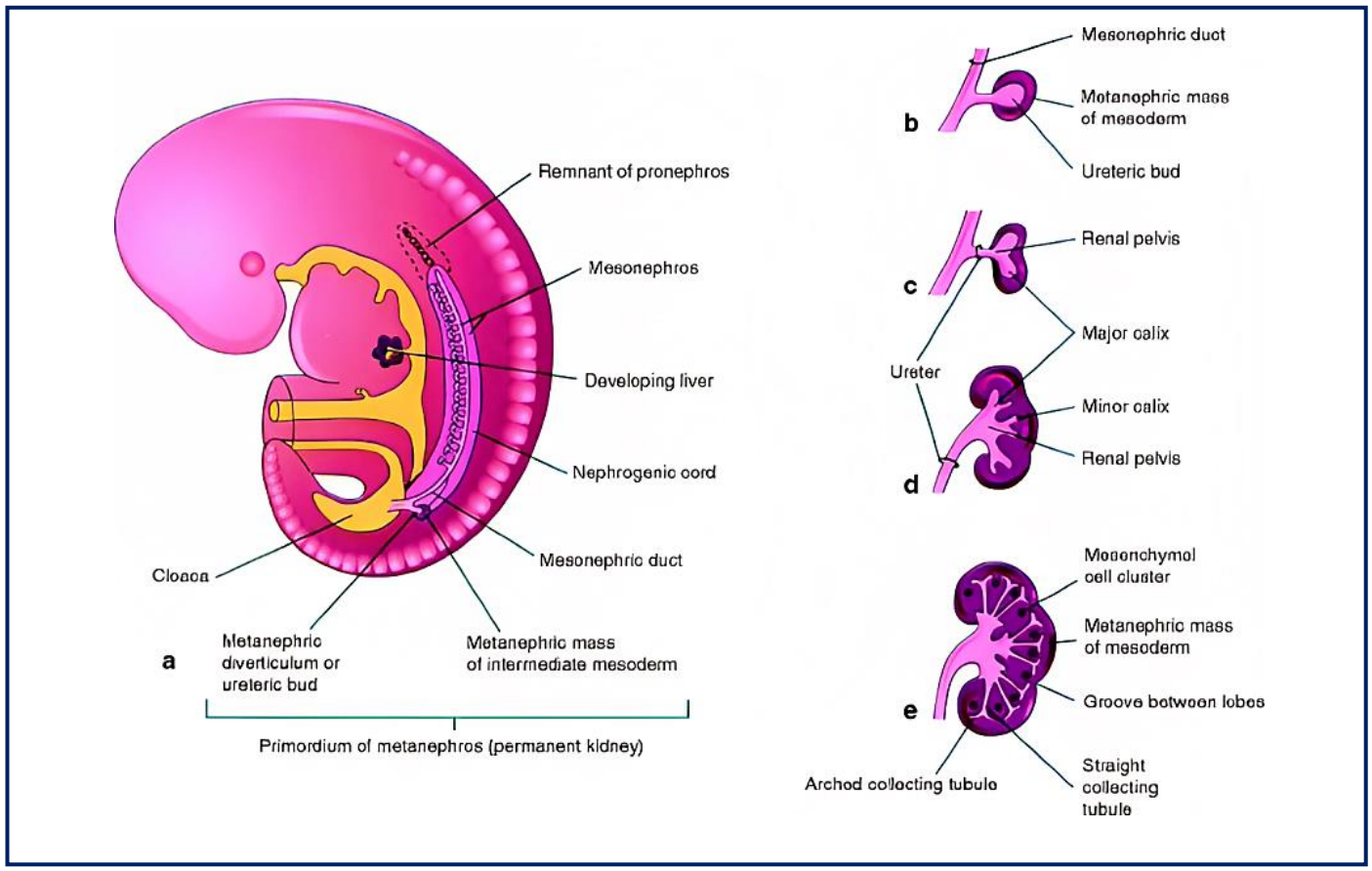
4.A DIVERTICULUM ARISES FROM THE MESONEPHRIC DUCT CALLED URETERIC BUD
↓
WHICH FORMS METANEPHRIC BLASTEMA.

THE KIDNEYS ARE DERIVED FROM TWO SOURCES: -

1. EXCRETING COMPONENT - DERIVED FROM METANEPHROS.
- METANEPHRIC BLASTEMA – CONTACT WITH AMPULLA → UNDERGO DIFFERENTIATION FORM NEPHRON

2. COLLECTING COMPONENT - DERIVED FROM URETERIC BUD.
- URETIC BUD GROW TOWARD METANEPHRIC BLASTEMA → BECOME AMPULLA
- AMPULLA → FORM PELVIS OF KIDNEY -> THEN FORM MINOR & MAJOR CALYX AND COLLECTING TUBULES

THE KIDNEYS BECOME FUNCTIONAL FROM 12TH WEEK.
ASCENT OF KIDNEY – INITIAL IN SACRAL REGION → KIDNEY ASCEND TO LUMBAR REGION
ROTATION OF KIDNEY – HILUM -> FIRST ANTERIOR, LATER HILUM COME MEDIALLY




## DEVELOPMENT OF ANAL CANAL

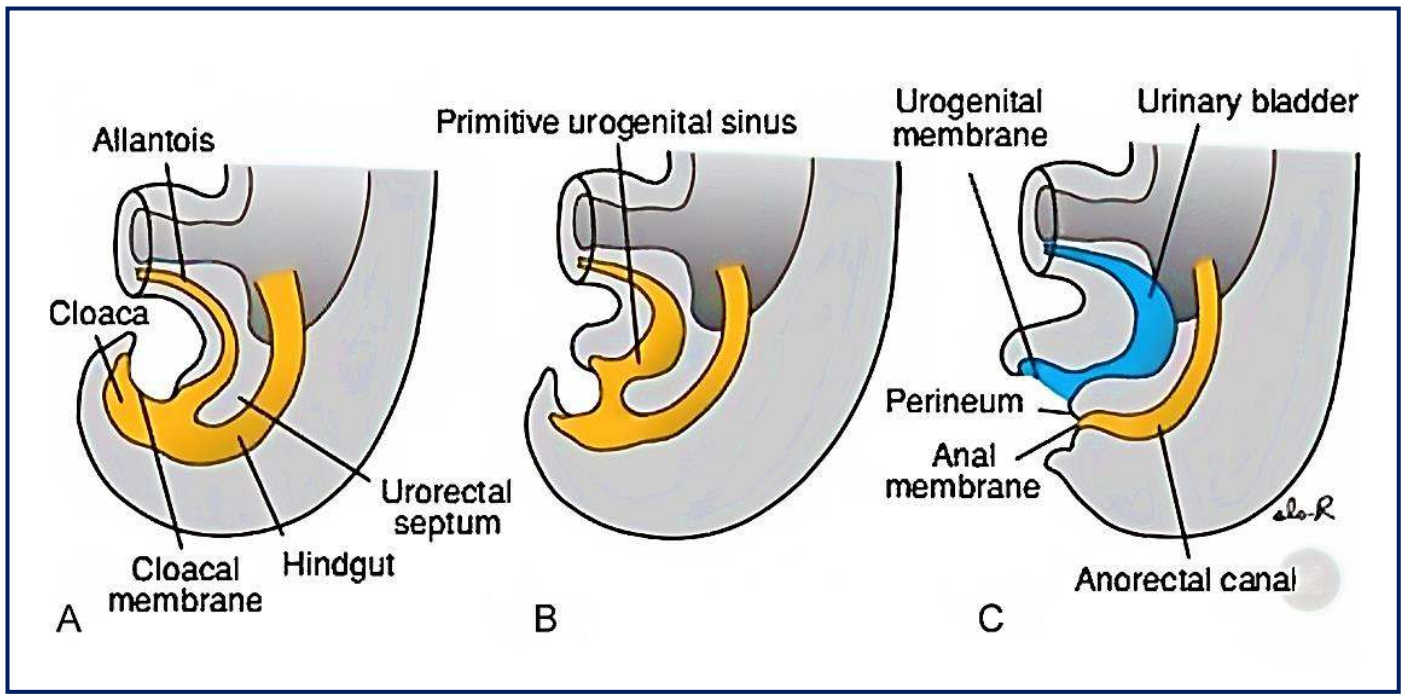
DEVELOPMENT OF HINDGUT

1. DIVERTICULUM ARISING FROM HINDGUT IS CALLED ALLANTOIS.
2. CLOACA IS A PART OF HINDGUT DISTAL TO ALLANTOIS.
3. THE URORECTAL SEPTUM -> SEPARATES THE CLOACA AND CLOACAL MEMBRANE.
4. THE CLOACA DIVIDES INTO -> PRIMITIVE UROGENITAL SINUS
-> PRIMITIVE RECTUM.
5. THE CLOACAL MEMBRANE DIVIDES INTO -> ANAL MEMBRANE & UROGENITAL MEMBRANE.
6. THE PRIMITIVE RECTUM FORMS -> RECTUM, ANAL CANAL UP TO PECTINATE LINE

DERIVATION OF UPPER ANAL CANAL:
- ENDODERM
↓
GUT → HINDGUT → CLOACA → PRIMITIVE RECTUM → UPPER PART OF ANAL CANAL

DERIVATION OF LOWER PART OF ANAL CANAL:
- ECTODERM → PROCTODEUM → LOWER PART OF ANAL CANAL

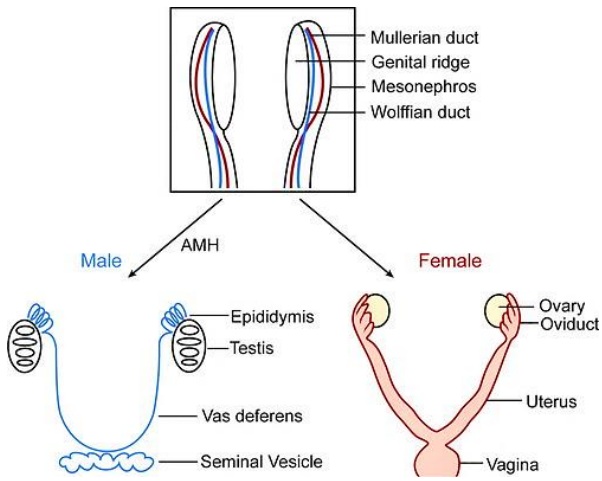
THE ANAL MEMBRANE IS FORMED AT THE JUNCTION OF UPPER AND LOWER ANAL CANAL.
IT RUPTURES BY THE 9TH WEEK THEREFORE; UPPER AND LOWER ANAL CANAL COMMUNICATE.
THE PECTINATE LINE IS FORMED AT THE SITE OF FORMATION OF ANAL MEMBRANE.




## DEVELOPEMNT OF GONADS

GENOTYPE THAT IS XX/XY IS DETERMINED AT THE TIME OF FERTILIZATION ITSELF

PHENOTYPIC DIFFERENTIATION WILL START ONLY AT 7TH WEEK
UNTIL 6TH WEEK GONADS ARE INDIFFERENT GONADS
PRIMORDIAL GERM CELLS REACH THE GONADS AT 5TH WEEK




GONADS

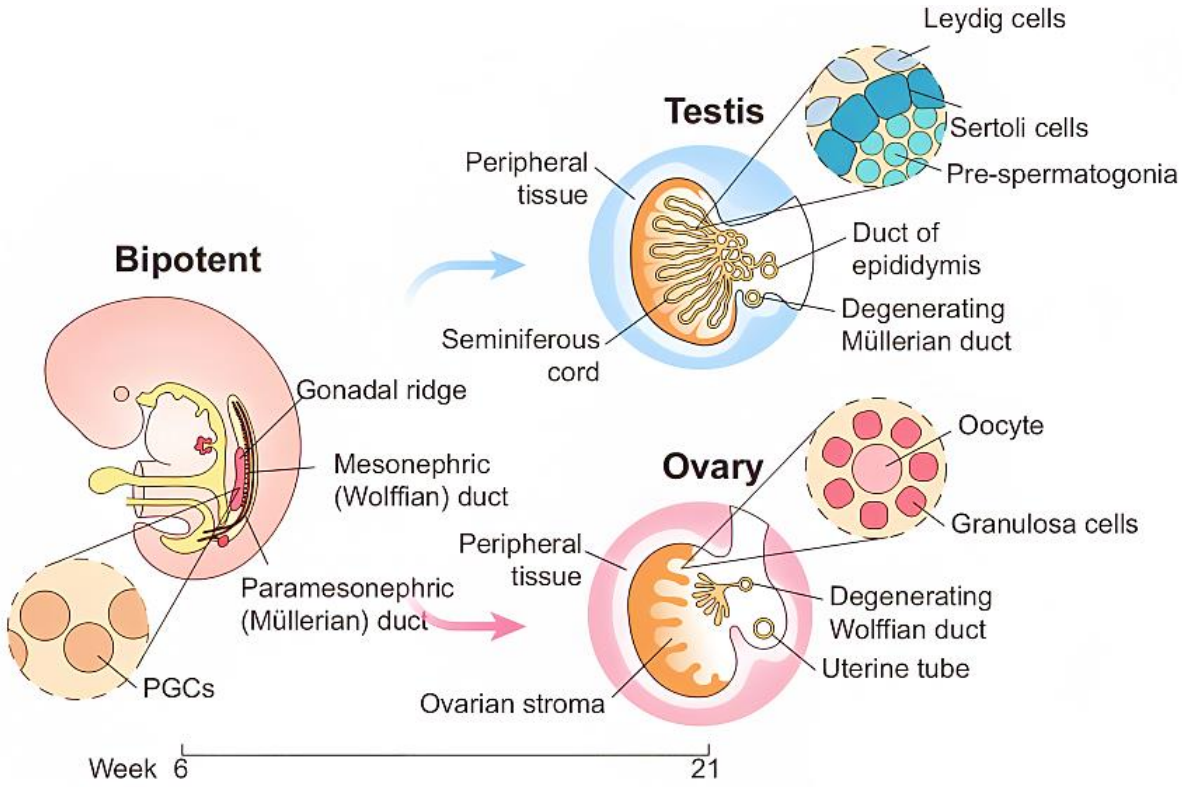
TESTIS DEVELOPS AT 7TH WEEK
OVARY DEVELOPS AT 8TH - 10TH WEEK

EXTERNAL GENITALIA:

- IN MALES: 14TH WEEK
- IN FEMALES: 11TH WEEK

SEX DETERMINATION CAN BE DONE AT 15TH WEEK (ILLEGAL TO REVEAL

| STRUCTURES | MALE DERIVATIVE | FEMALE DERIVATIVE |
|---|---|---|
| 1. GONADAL RIDGE | - TESTIS | - OVARY |
| 2. GENITAL TUBERCLE | - PENIS | - CLITORIS |
| 3. GENITAL SWELLING | - SCROTUM | - LABIA MAJORA |
| 4. MESONEPHROS | - EFFERENT TUBULES | - EPOOPHORON |
| 5. MESONEPHRIC DUCT | - APPENDIX OF EPIDIDYMIS<br>- VAS DEFERENS<br>- EPIDIDYMIS<br>- EJACULATORY DUCT<br>- SEMINAL VESICLE | - APPENDIX OF OVARY |
| 6. PARAMESONEPHRIC DUCT | - APPENDIX OF TESTIS | - FALLOPIAN TUBE<br>- UTERUS<br>- VAGINA (UPPER PART) |




## DESCENT OF TESTIS

1) DESCEND TO REACH SCROTUM.
2) EACH TESTIS BEGIN TO DESCENT DURING SECOND MONTH OF INTRAUTERINE LIFE
3) REACH ILIAC FOSSA BY 3RD MONTH
   - DEEP INGUINAL RING - 4TH TO 6TH MONTH
   - INGUINAL CANAL 7TH MONTH
   - SUPERFICIAL INGUINAL RING → 8TH MONTH
   - BOTTOM OF SCROTUM BY 9TH MONTH

4) EXTENSION OF PERITONEAL CAVITY -> CALLED PROCESSUS VAGINALIS →PRECEDES THE DESCENT OF TESTS INTO THE SCROTUM
5) PROCESSUS VAGINALIS CLOSE ABOVE THE TESTIS
6) DESCENT DOESN'T OCCUR AFTER ONE YEAR OF AGE